FACULTY OF ENGINEERING

Engineering Design 2026

Group 13 report

By:

RN Nel 40496325

VA King 32504462

F Prinsloo 44415273

AR Cronje 50536540

MM Van Rooyen 30300649

Submitted in pursuit of the degree

BACHELOR OF ENGINEERING

North-West -University Potchefstroom Campus

2026

Table of Contents

1 Overview 1

1.1 Allocation of Work 1

2 Interface Control 2

2.1 Document Control 2

2.2 Purpose and Scope 3

2.3 System Overview 3

2.4 Interface Inventory 5

2.5 Detailed Interface Specifications 7

3 Main Design Decisions 28

3.1 MCTR 1 – Lateral 28

3.2 MCTR 2- Spooling 29

3.3 EEII 1 – Drive circuit 30

3.4 EEII 2 – Magnet 31

3.5 REII – Remote Control Dashboard and RF Ground Station 33

4 Integration 35

4.1 Pre-integration Test Plan 35

4.2 Overall performance test plan 38

5 Progress 41

5.1 MCTR 1 – Lateral Motion 41

5.2 MCTR 2 - Spooling 42

5.3 EEII 1 – Drive circuit 43

5.4 EEII 2 – Magnet 45

5.5 REII- Remote Control Dashboard and RF Ground Station 47

Appendix A 51

List of Figures

Figure 1: System Block diagram. 4
Figure 2: IF1.1 MCTR1 side connector terminals. 8
Figure 3: IF1.2 Motor encoder connector. 8
Figure 4: MCTR1 motor testing rig as Implementation evidence. 41
Figure 5: Progress of Gearbox and spooling system. 42
Figure 6: S type load cell 42
Figure 7: Low voltage single phase switching. 43
Figure 8: Drive circuit schematic in Kicad 43
Figure 9: Drive circuit PCB layout. 44
Figure 10: 3D model of drive circuit PCB. 44
Figure 11: Core details 45
Figure 12: Electromagnet circuit layout. 45
Figure 13: Magnet switching circuit testing. 46
Figure 14: Magnet housing design 46
Figure 15: Magnet housing top view 47
Figure 16: Bench test setup of the ESP32-S3 crane System Controller (FU8, COM12) and prototype NodeMCU-32S USB ground-station radio unit (FU6, COM10) during Milestone 4 two-way RF verification. 49
Figure 17: Measured 200-packet wireless smoke test results (smoke.csv, Run 20261016) on Channel 77 (2477 MHz) at 2 Mbps: (a) per-round RTT and Auto-ACK status; (b) latency and outcome distribution (0.89 ms mean RTT); and (c) cumulative crane rig reception (200/200) vs. ground-station Auto-ACK confirmation (192/200) 50

List of Tables

Table 1: Work Division 1
Table 2: Document admin 2
Table 3: Document control 2
Table 4: Scope of work 3
Table 5: Functional unit descriptions 5
Table 6: Interface inventory 5
Table 7: IF1.0 description 7
Table 8: IF1.1 Physical Description 7
Table 9: IF1.2 Physical Description 8
Table 10: IF1.1 Electrical Description 9

Table 11: IF1.2 Electrical Description 9
Table 12: IF1.1 Pin definitions 9
Table 13: IF1.2 Pin definitions 10
Table 14: IF1.1 Communication description 10
Table 15: IF1.2 Communication description 11
Table 16: IF6.0 description 12
Table 17: IF6.0 Physical Description 12
Table 18: IF6.0 Electrical Description 12
Table 19: IF6.0 Pin definitions (Main Board Headers J3, J6 & J7) 13
Table 20: IF6.0 Communication description 14
Table 21: IF7.0 description 15
Table 22: IF7.0 Physical Description 15
Table 23: IF12.0 description 16
Table 24: IF12.0 Physical Description 16
Table 25: IF12.0 Electrical Description 17
Table 26: IF12.0 Pin definitions 17
Table 27: IF12.0 Communication description 18
Table 28: : IF10.0 description 19
Table 29: IF10.0 Physical Description 19
Table 30: IF10.0 Electrical Description 20
Table 31: IF10.0 Pin definitions 21
Table 32: IF10.0 Communication description 21
Table 33: IF10.0 Downlink command payload map 22
Table 34: IF10.0 Uplink telemetry ACK-payload map 23
Table 35: IF13.0 & IF15.0 description 24
Table 36: IF13.0 & IF15.0 Physical Description 25
Table 37: IF13.0 & IF15.0 Electrical Description 25
Table 38: IF13.0 & IF15.0 Pinout Description 26
Table 39: IF13.0 & IF15.0 Communication Description 27
Table 40: Carriage system design decisions 28
Table 41: Magnet design choices 32
Table 42: Core dimensions summary 32
Table 43: Overall test plan and performance verification 40
Table 44: REII milestones and implementation evidence. 47
Table 45: Acronyms and abbreviations 51

1 Overview

The project entails the design and implementation of a hoist system. The system was divided into subsystems namely Lateral movement, Spooling, Drive electronics, Magnetic attachment and Remote control. The subsystems were allocated to the following engineering disciplines within the group.

1.1 Allocation of Work

The work was divided into five work packages as in the following table.

Table 1: Work Division

Name	Student Nr	Work package
R Nel	40496325	REII
VA King	32504462	MCTR2
F Prinsloo	44415273	EEII2
AR Cronje	50536540	MCTR1
MM Van Rooyen	30300649	EEII1

The following subsections provide a brief introduction on the scope of work allocated to each member of the group.

1.1.1 MCTR 1 – Lateral Movement

Design and assemble a driven carriage that allows the system to be propelled along the given rail. This includes two DC motors attached to a system of gears and wheels that fit on the rail. The motor is controlled indirectly by the MCU (external to the subsystem) with two dedicated motor controllers and each motor is fitted with shaft encoders that measure rotation which is sent back as a digital signal to the MCU. All other subsystems need to mount onto the carriage.

1.1.2 MCTR 2 - Spooling

Develop, design and manufacture a hoist gearbox capable of lifting a maximum load of 20 kg. The gearbox uses a speed-reduction system with a combination of six gear stages, and a worm gear configuration is excluded by design constraint. It also includes a fail-safe active brake that engages automatically on loss of drive power or command signal and holds the load in place. An on-hook load measurement is provided by a load cell in the hoist path, which feeds both telemetry and the safety interlocks.

1.1.3 EEII 1 – Electronic Drive circuit

Design and manufacture an electronic drive circuit to control the three-phase induction motor, responsible for vertical movement(spooling). The drive circuit includes protection against fault currents

and protection of the microcontroller of the system. The distribution of 220V AC within the system was also assigned to this discipline.

1.1.4 EEII 2 – Magnet Load attachment

Design, construction, and implementation of an electromagnet. The electromagnet was theoretically designed and simulated; once the magnet had successful simulation results, the electrical circuit was constructed to achieve these results. Using a MOSFET switching module and a secondary power supply, the magnet is energised and used to magnetically attach the payload to the hoisting system. The switching module allows for control over the voltages seen by the electromagnets and regulates the current generated.

1.1.5 REII – Remote Control Dashboard and RF Ground Station

Design and implement a wireless remote-control dashboard and ground-station radio link to control the crane and monitor live sensor readings from a PC. A ground-station microcontroller plugs into the operator PC via USB and uses a 2.4 GHz nRF24L01+ wireless module to communicate with the main ESP32-S3 controller on the crane using fixed 32-byte data packets. This subsystem also includes smooth motor acceleration control (anti-sway) to reduce load swing, automated path planning around obstacles, and wireless range and reliability testing.

2 Interface Control

Each of the above subsystems need to function together and thus introduces many interfaces between different members. The following section attempts to document the interface agreements.

Table 2: Document admin

Document field	Details
Document number	ICD-001
Revision	Rev. A
Date	2026-10-23
Prepared by	Group 13
Approved by	Group 13
Document status	Initial release

2.1 Document Control

Table 3: Document control

Revision	Date	Description of change	Author
A	2026-10-24	Initial release	Group 13

2.2 Purpose and Scope

2.2.1 Purpose

This section serves as the initial Interface Control Document (ICD) which defines the technical and functional requirements for the interfaces between relevant subsystems. It establishes the physical, electrical, mechanical and/or communication requirements needed for compatibility and correct operation of the integrated system.

2.2.2 Scope

Table 4: Scope of work

Item	Description
1. Included	Components, subsystems, connectors, signals and communication links covered by this ICD.
1.1 Subsystems	Power distribution, Drive electronics, Hoisting system, Electromagnetic attachment, On-board system control, Remote control & Ground-station, Lateral motion.
1.2 Components	Breaker(s), power supplies, VFD controller, Induction motor, Hoist Gearbox, Spool, Fail-safe hoist brake, Electromagnets, MOSFET switching circuit, Load cell, Crane MCU, Ground-station MCU, dual 2.4 GHz radio modules, PC operator dashboard, DC motor drivers, encoders, DC motors, carriage.
1.3 Connectors	PSU cables, PSU terminals, PWM signal cables (VFD + DC motor driver + Magnet), Digital control cables (Direction, Brake, VFD Enable/Fault), Load cell and current sensor cables, Hoist cable/rope, encoder cables, Physical mounting connections, Induction motor terminals, MCU carrier board headers, USB cable (PC to ground station).
1.4 Signal/Communication	MCU -> VFD, VFD -> Induction motor, MCU -> DC motor controller, Encoder -> MCU, Load cell -> MCU, MCU -> Hoist brake, MCU -> Magnet controller, Current sensors -> MCU, PC Dashboard -> Ground-station radio, Ground-station radio -> Crane radio, Crane radio -> Crane MCU.
2. Excluded	Interfaces or functions not covered.
2.1 Interfaces/Functions	Internal mechanical/electrical sub-assemblies within a single work package (Induction motor -> Gearbox, Gearbox -> Spool, Magnets -> Magnet frame, Magnet controller -> Coils, DC motor controller -> DC motors) and internal PC operating-system software. As well as all interfaces with the electrical power distribution subsystem, as this is left for future versions of the ICD's. Note: The PC operator dashboard, USB ground-station link, and 2.4 GHz wireless link are included in this ICD.

2.3 System Overview

The following section is dedicated to the interconnection of subsystems. The relevant interfaces between subsystems are represented in the system block diagram in this section.

2.3.1 System Block Diagram

The system block diagram defines the interaction between subsystems in the project. The system is confided within the dashed block. The components on the outside refer to interfaces not included in the system itself, but rater interaction of the system with the environment it operates in. The subsystems were defined in terms of functional unit blocks. The following system block diagram was created for the hoist system:

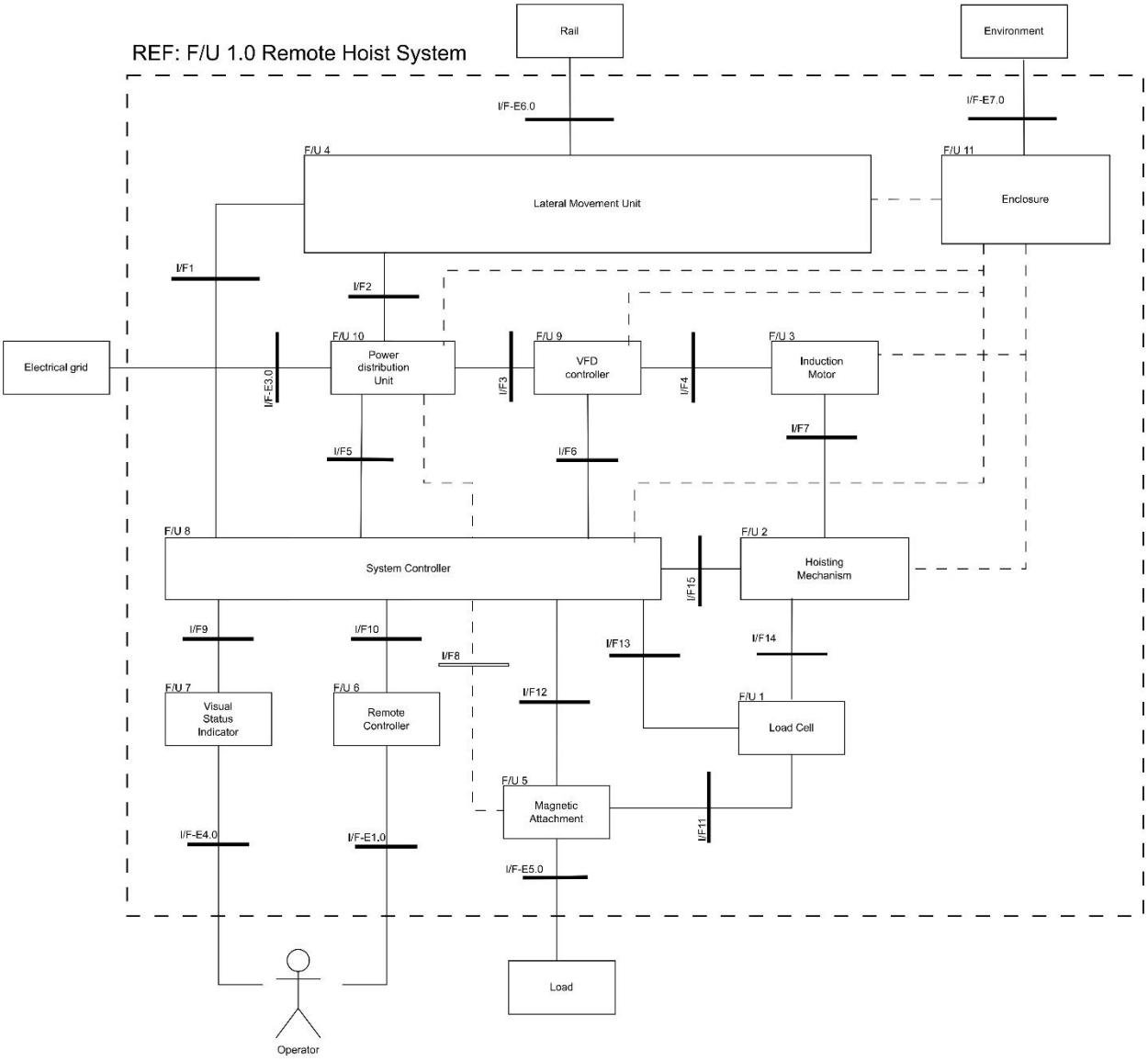


Figure 1: System Block diagram.

The functional unit blocks were connected by solid lines to indicated interfaces associated with each subsystem as seen in Figure 1. The functional unit blocks were summarized in the following section.

2.3.2 Functional Unit Descriptions

Table 5: Functional unit descriptions

Subsystem ID	Name	Function	Responsible party
FU1	Load cell	Sensor unit (S-type) that is connected between the cable and magnetic attachment that measures onboard loads through tension and feeds the information back to an ADC pin of the controller.	MCTR2
FU2	Hoist mechanism	Main mechanical assembly that houses the gearbox and spool system. The gearbox output drives the spool, and the brake acts on the same drivetrain.	MCTR2
FU3	Induction motor	Three phase induction motor supplying torque to the hoist mechanism to drive gearbox of spooling system.	MCTR2 & EEII1
FU4	Motorized carriage	Acts as the connection between system and rail and controls lateral motion and motion sensing.	MCTR1
FU5	Magnetic Attachment	Electromagnet used to attach and release the payload from the hoist.	EEII2
FU6	Remote Controller	PC operator dashboard and USB ground-station radio unit that sends movement and magnet commands wirelessly to the crane, calculates smooth anti-sway speed ramps, and displays live sensor readings and fault alerts.	REII
FU7	Visual Status Indicator	On-board RGB status light (LED D4 on the main controller board) that shows the crane's operating state and fault conditions visually.	REII
FU8 (Shared)	System Controller	Central ESP32-S3 microcontroller board mounted on the crane that receives wireless commands, reads all onboard sensors, enforces safety limits, and outputs control signals to the motors, VFD, brake, and electromagnet.	Shared (All Disciplines)
FU9	VFD controller	Circuit converting single phase AC supply to three phase AC in order to drive the 3-phase induction motor as controlled by the switching sequence PWM signals from the MCU.	EEII1
FU10	Power distribution unit	Distribution board with single phase AC incoming from which power supplies of all subsystems are energized.	EEII1
FU11	Frame/Enclosure	Frame onto which all subsystems mount and possible enclosure that protect from environment.	MCTR1 & MCTR2

2.4 Interface Inventory

The interfaces defined in Figure 1 was describes in the following table:

Table 6: Interface inventory

Interface ID	Connected	Interface type	Description / drawing reference
Internal interfaces			
IF1.0	FU4 ↔ FU8	Electrical (PWM & Digital)	Speed PWM and direction signals from the System Controller to the lateral DC motor driver, plus wheel encoder pulses back to the controller.

IF2.0	FU4 ↔ FU10	Electrical (Power)	Power supply connection from the Power Distribution Unit to the lateral movement motor driver.
IF3.0	FU9 ↔ FU10	Electrical (Power)	Single-phase 220 V AC power connection from the Power Distribution Unit to the VFD controller.
IF4.0	FU3 ↔ FU9	Electrical (3-Phase Power)	Variable-frequency three-phase AC power output from the VFD controller to the hoisting three phase induction motor.
IF5.0	FU8 ↔ FU10	Electrical (Power)	Stepped-down DC power supply connection from the Power Distribution Unit to the System Controller board.
IF6.0	FU8 ↔ FU9	Electrical (PWM, Digital & Analog)	Three-phase 20 kHz PWM control signals and enable/fault logic between the System Controller and VFD, plus drive current feedback.
IF7.0	FU2 ↔ FU3	Mechanical	Mechanical shaft coupling between the induction motor output shaft and the hoisting gearbox input stage.
IF8.0	FU5 ↔ FU10	Electrical (Power)	Single phase AC power connection for magnetic attachment power supply.
IF9.0	FU7 ↔ FU8	Internal PCB Trace	Built-in board trace connecting the System Controller to the on-board RGB status LED (D4) (internal to FU8; excluded from external subsystem ICDs).
IF10.0	FU6 ↔ FU8	Wireless RF & Digital Serial	2.4 GHz wireless link (Channel 77) between the Remote Controller ground station and the crane System Controller for commands and live telemetry.
IF11.0	FU1 ↔ FU5	Mechanical	Mechanical attachment coupling the bottom of the S-type load cell to the magnetic load attachment frame.
IF12.0	FU5 ↔ FU8	Electrical (PWM, Digital & Analog)	PWM coil control signal and enable logic from the System Controller to the electromagnet switch module, plus coil current feedback (commanded via IF10.0).
IF13.0	FU1 ↔ FU8	Electrical + Differential Analog	Millivolt tension signal and 3.3 V power excitation between the S-type load cell and the System Controller's 24-bit analog-to-digital converter.
IF14.0	FU1 ↔ FU2	Mechanical	Mechanical coupling between the steel hoist cable and the top of the S-type load cell.
IF15.0	FU2 ↔ FU8	Electrical (Digital Control)	Digital ON/OFF control signal from the System Controller to the hoisting mechanism’s fail-safe electromagnetic brake module.

Please note that interface between functional units and the enclosure were not included in this summary. The following section exhibit the most important interphases in detail.

2.5 Detailed Interface Specifications

The following subsection was dedicated to detailed interface control agreements (only main interfaces between subsystems included for this version).

2.5.1 Interface IF1.0

Interface between FU 8 (System Controller) and FU 4 (Lateral Movement Unit).

Table 7: IF1.0 description

Field	Specification
Description / purpose	This interface controls the speed and direction command sent to the motor controllers and reads the encoder positions.
Connected components:	
IF1.1	MCU ↔ DC Motor controllers
IF1.2	Quadrature Encoders ↔ MCU
Interface owner(s)	MCU side – REII, Motor & Encoder side – MCTR1

Physical / Mechanical Interface

IF1.1 (MCU ↔ Motor controllers)

Table 8: IF1.1 Physical Description

Parameter	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	Standard 2.54 mm pitch Dupont jumper connectors between the MCU carrier board headers and the HW-095 (L298N) motor driver pins.	Standard 2.54 mm pin pitch
Connector gender / orientation	Control lines: Female-to-Female Dupont jumpers. Power Supply lines: Male (MCTR 1) to Female (REII).	Verify +5 V and GND polarity before plugging in
Pin count (per controller)	5 active connections (1 PWM speed wire, 2 direction control wires, +5 V logic supply, and common GND; plus 2 optional direction wires if the second H-bridge channel is driven separately).	5-7 pins total
Cable type	24–26 AWG ribbon cable or isolated stranded jumper wire.	≤ 300 mm length to avoid PWM signal degradation
Connector mounting	See **Error! Reference source not found.** f or illustration of mounting terminals.	-

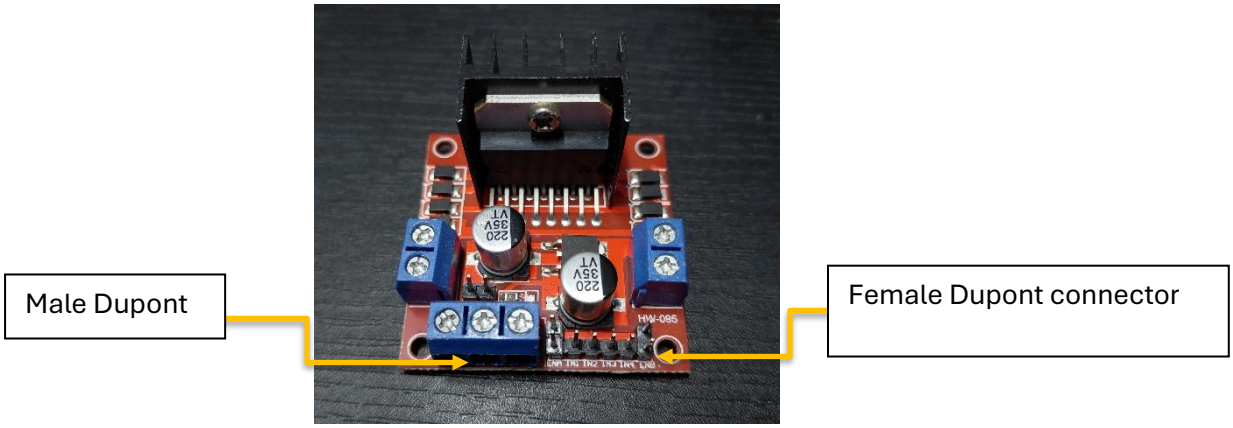


Figure 2: IF1.1 MCTR1 side connector terminals.

IF1.2 (Quadrature Encoder ↔ MCU)

Table 9: IF1.2 Physical Description

Parameter	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	Standard Dupont jumper connectors from the motor encoder harness to the MCU carrier board (encoder mounted on DC motor rear shaft).	Standard 2.54 mm pin pitch
Connector gender / orientation	Male jumper pins (MCTR1 encoder side) to Female header socket (MCU side).	-
Pin count (per encoder)	3 active connections (1 encoder pulse signal wire, +5 V supply, common GND; optional 2nd pulse wire if two-channel quadrature is used).	3-4 pins total
Cable / harness type	24–26 AWG stranded wire (twisted pair with GND recommended)	≤ 300 mm length to avoid noise on encoder pulses
Connector mounting	See Figure 3 for illustration of cable from encoder.	-

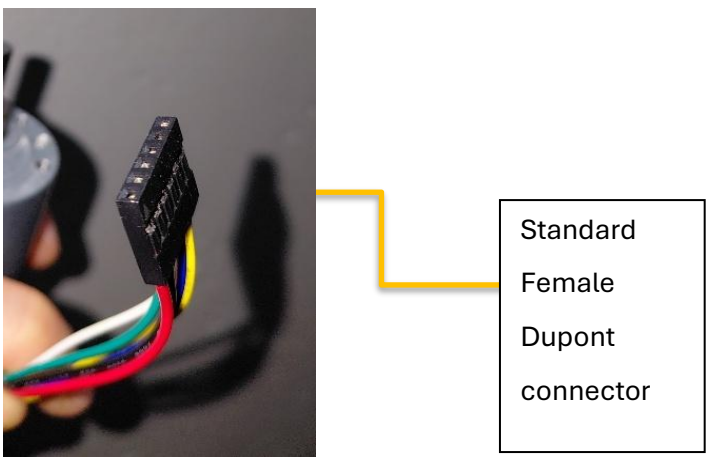


Figure 3: IF1.2 Motor encoder connector.

Electrical Interface

IF1.1 (MCU ↔ Motor controllers)

Table 10: IF1.1 Electrical Description

Parameter	Nominal requirement	Minimum / maximum / tolerance
Supply voltage	5 V DC logic power supplied from the MCU carrier board to the HW-095 motor driver logic circuit.	6 V max
Current	0 – 36 mA (internal logic draw)	MCU / MCP23017 max sink/source: ~20–25 mA per digital control pin
Signal type (per controller)	1 PWM speed signal and 2 digital HIGH/LOW direction signals.	PWM frequency: 1–10 kHz
Logic levels	3.3 V digital logic output from the MCU carrier board (LOW = 0.0 V, HIGH = 3.3 V).	HW-095 accepts 2.3 to 5.0 V as logic HIGH.
Ground / reference	Common digital ground shared between the MCU carrier board and the DC motor driver.	Must share common ground to prevent false triggering

IF1.2 (FU10 ↔ FU4)

Table 11: IF1.2 Electrical Description

Parameter	Nominal requirement	Minimum / maximum / tolerance
Supply voltage	5 V DC supply from the MCU carrier board to power the encoder.	4.75 V to 5.25 V DC
Current	Low current (~10-20 mA per encoder)	-
Signal type	Digital square-wave pulse train.	Max pulse voltage at MCU pin must not exceed 3.3 V
Logic levels	3.3 V logic at the MCU input (LOW: $0 - 0.8$ V, HIGH: $2 - 3.3$ V).	Note: Because the encoder is powered from 5 V, a voltage divider or level shifter to must be used to step the 5 V pulse down to 3.3 V.
Ground / reference	Common digital ground shared with the MCU carrier board.	-

Pinout and Signal Definitions

IF1.1(MCU ↔ Motor controllers)

Table 12: IF1.1 Pin definitions

Pin / terminal	Signal name	Direction (relative to MCU)	Electrical characteristics	Function / notes
J4.2 (or J9.1)	VCC (+5V)	Output (Rail)	5.0 V DC Supply	5 V logic power supply to HW-095 motor driver & encoder

J4.1 (or J9.2)	GND	Reference	0 V Digital GND	Common ground reference return.
J7.9 (Silkscreen A)	LAT_PWM_ENA	Output	3.3 V PWM (1–10 kHz)	Lateral Motor Speed PWM connected to HW-095 ENA pin.
J4.4 (EXP_PortB GPB1)	MOT_IN1	Output	3.3 V Digital Logic	Motor Direction 1 control connected to HW-095 IN1 pin.
J4.5 (EXP_PortB GPB2)	MOT_IN2	Output	3.3 V Digital Logic	Motor Direction 2 control connected to HW-095 IN2 pin.
J4.6 (EXP_PortB GPB3)	MOT_IN3 (Optional)	Output	3.3 V Digital Logic	Optional Motor B Direction 3 (if 2nd H-bridge channel wired)
J4.7 (EXP_PortB GPB4)	MOT_IN4 (Optional)	Output	3.3 V Digital Logic	Optional Motor B Direction 4 (if 2nd H-bridge channel wired)

IF1.2(FU10 ↔ FU4)

Table 13: IF1.2 Pin definitions

Pin / terminal	Signal name	Direction (relative to MCU)	Electrical characteristics	Function / notes
J4.2 (or J9.1)	VCC (+5V)	Output (Power Rail)	5.0 V DC Supply	5 V supply to motor shaft encoder.
J4.1 (or J9.2)	GND	Reference	0 V Digital GND	Common ground reference return
J7.12 (Silkscreen D)	ENC_A	Input	3.3 V Pulse Train	Wheel Encoder Channel A (1050 counts per wheel turn; stepped down from 5 V to 3.3 V)
J7.11 (Silkscreen C)	ENC_B (Optional)	Input / Unwired	3.3 V Pulse Train	Optional Encoder Channel B (or left unconnected for single-channel counting)

Data / Communication Interface

IF1.1 (MCU ↔ Motor controllers)

Table 14: IF1.1 Communication description

Parameter	Specification
Data type	PWM speed signal + digital HIGH/LOW direction lines
Frequency	1-10kHz PWM

Input-Output relation (Linear)	0% Duty Cycle -> 0 RPM (stopped); 100% Duty Cycle -> 380 RPM (maximum speed); IN1/IN2 set forward, reverse or brake.

IF1.2 (FU10 ↔ FU4)

Table 15: IF1.2 Communication description

Parameter	Specification
Data type	Digital square-wave encoder pulses
Counts per revolution	700 counts per motor shaft revolution / 1050 counts per wheel revolution (after 1.5:1 gear stage)
Protocol	(Number of rising edges on A or B)*1050 = wheel revolutions Travel direction is known from the commanded motor direction (IN1/IN2).

Interface Dependencies and Constraints

REII specific:

- **Dedicated PWM & Direction Pins**: Motor speed is controlled by a smooth hardware PWM signal on pin J7.9, while the forward/reverse direction signals (IN1 and IN2) are placed on the digital output header (pins J4.4 and J4.5) so they do not use up high-speed timer pins.
- **Hardware Pulse Counting**: The microcontroller uses its built-in pulse counter on pin J7.12 (and J7.11 if Channel B is wired) to count wheel encoder ticks automatically in hardware, ensuring no position counts are missed, while wireless packets are being processed.
- **3.3 V Pin Protection**: The HW-095 motor driver accepts the board's 3.3 V logic signals directly, but the 5 V encoder pulse must pass through a simple resistor voltage divider (or level shifter) to step down the 3.3 V before reaching pin J7.12.

MCTR1 specific:

- **Pin constraints on MCU:** To reduce pin consumption a single motor controller is used in such a way that one motor control input drives both motors. Each motor will still have separate output terminals, but the control signals will be shared between ENA and ENB as well as between IN1, IN2 and IN3,IN4. This can be done as the motors will always rotate in the same direction at the same speed. Additionally, a single encoder will be used with a single channel output, as the second encoder is redundant. The accuracy of the encoder reading is not of utmost importance and therefore a single encoder can be used. A single encoder channel is sufficient because the direction can be determined from within the code.

2.5.2 Interface IF6.0

Interface between FU 8 (System Controller) and FU 9 (VFD Controller)

Table 16: IF6.0 description

Field	Specification
Description / purpose	Supplies three-phase 20 kHz sinusoidal PWM (SPWM) switching signals and drive enable/fault logic from the System Controller to the VFD circuit and returns an analog current measurement from the drive circuit.
Connected components	MCU ↔ VFD
Interface owner(s)	MCU side – REII, VFD side – EEII 1

Physical / Mechanical Interface

Table 17: IF6.0 Physical Description

Parameter	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	Standard 2.54 mm pitch Dupont jumper connectors on the MCU carrier board to terminal/header connectors on the VFD gate-driver and current-sensor board.	Standard 2.54 mm pitch
Connector gender / orientation	Female Dupont sockets on the MCU board headers to male header pins on the VFD control board.	Verify phase order (Phases A, B, C) and ground orientation
Pin / contact count	8 connections total: 3 PWM phase outputs, 1 drive enable output, 1 fault status input, 1 digital ground, 1 analog current sensor input, and 1 analog ground reference.	8 wires total
Cable / harness type	24–26 AWG insulated stranded copper wire; keep low-voltage signal harness routed away from 220 V AC motor power leads.	≤ 300 mm length to minimize switching noise pickup
Mounting arrangement	MCU board and VFD board mounted inside the crane enclosure with optical isolation separating the 220 V power stage from the 3.3 V controller.	-

Electrical Interface

Table 18: IF6.0 Electrical Description

Parameter	Nominal requirement	Minimum / maximum / tolerance
Supply voltage	3.3 V DC peripheral supply rail from the MCU carrier board for low-voltage logic and current sensor reference.	3.0 V to 3.6 V DC

Current	5 – 15 mA per opto-coupler LED input; < 5 mA on digital enable/fault and analog current sense lines.	Max 20 mA per MCU output pin
Signal type (per controller)	3x 20 kHz sinusoidal PWM outputs (120° apart), 2x digital HIGH/LOW lines (Drive Enable & Fault status), and 1x low-voltage analog current signal.	20 kHz PWM carrier; 0 – 3.3 V analog current signal
Logic levels	3.3 V digital logic (LOW = 0.0 V, HIGH = 3.3 V).	Compatible with 4N25X opto-coupler (3.3V or 5V)
Ground / reference	MCU digital ground connected only to the input (LED) side of the VFD opto-couplers; analog ground reference used for the current sensor.	High-voltage VFD ground must remain isolated from MCU ground.

Pinout and Signal Definitions

IF6.0(MCU ↔ VFD)

Table 19: IF6.0 Pin definitions (Main Board Headers J3, J6 & J7)

Pin / terminal	Signal name	Direction (relative to MCU)	Electrical characteristics	Function / notes
J7.8 (Silkscreen 1 / IO1)	VFD_SPWM_A	Output	3.3 V PWM (20 kHz)	Three-phase inverter SPWM Phase A (0° reference) to VFD opto-coupler
J7.7 (Silkscreen 2 / IO2)	VFD_SPWM_B	Output	3.3 V PWM (20 kHz)	Three-phase inverter SPWM Phase B (120° phase shift) to VFD opto-coupler
J7.6 (Silkscreen 3 / IO3)	VFD_SPWM_C	Output	3.3 V PWM (20 kHz)	Three-phase inverter SPWM Phase C (240° phase shift) to VFD opto-coupler
J3.4 (EXP_PortA GPA1)	DRV_EN_LOGIC	Output	3.3 V Digital Logic	Drive Enable control signal (HIGH = enable VFD output, LOW = inhibit)
J3.5 (EXP_PortA GPA2)	DRV_FAULT_IN	Input	3.3 V Digital Logic	Overcurrent / fault trip status signal from VFD protection circuit to MCU
J3.12 (Bottom of J3)	GND	Reference	0 V Digital GND	Common digital ground return for VFD opto-coupler LED side and logic

J6.7 (ADC_Port AN2)	VFD_CURR_IN	Analog Input	0 – 3.3 V Analog	VFD drive current sensor signal into on-board 24-bit ADC (Channel AN2)
J6.4 (ADC_Port GND)	GNDREF	Reference	0 V Analog GND	Filtered low-noise analog ground return for VFD current sensor
J6.12 (Bottom of J6)	+3.3VP	Output (Power Rail)	3.3 V DC Supply	3.3 V analog supply rail for current sensor conditioning circuit (if required)

Data / Communication Interface

IF1.1(MCU ⇔ VFD)

Table 20: IF6.0 Communication description

Parameter	Specification
Data type	3-Phase Sinusoidal PWM (SPWM), digital enable/fault status, and analog current feedback
Frequency	20 kHz PWM carrier modulated with 0 – 50 Hz three-phase sine wave (120° phase separation)
Input-Output relation (Linear)	Modulation frequency and amplitude set hoist motor speed; swapping two phases reverses motor direction (Up / Down); analog voltage on J6.7 scales linearly with drive current (reported in 0.1 A steps).

Operating Sequence

1. 3 PWM signals are applied to microcontroller output pins (120° out of phase)
2. 3 PWM signals received by VFD opto-couplers.
3. Isolated PWM signals used as input on VFD driver IC's.
4. Driver High and Low output control power electronics to achieve switching
5. Induction motor is powered by three phase AC power
6. Operate until MCU PWM outputs are terminated.

Interface Dependencies and Constraints

- The drive circuit is dependent on the MCU to ensure the phase shifts on the PWM signals are completely out of phase. If not, the power electronic IGBT's can switch on simultaneously on a per phase basis which could cause overcurrent and damage to components.
- The drive circuit is dependent on the weight of the load applied to the magnetic attachment. The weight directly influences the current drawn by the induction motor from the VFD. Overload can also cause overcurrent and possible damage to components.

2.5.3 Interface IF7.0

Interface between FU 3 (Induction motor) and FU 2 (Hoisting mechanism)

Table 21: IF7.0 description

Field	Specification
Description / purpose	Transfer torque and rotation from the induction motor output shaft to the bevel input stage of the hoist gearbox, which drives the spool for lifting and lowering
Connected components	Induction motor output shaft ↔ bevel pinion / gearbox input stage
Interface owner(s)	Mechanical coupling and motor mounting: MCTR 2 Motor electrical drive and speed control: EEII 1

Physical / Mechanical Interface

Table 22: IF7.0 Physical Description

Parameter	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	Bevel pinion mounted directly on the motor output shaft, secured with grub screw	Pinion bore matched to the shaft diameter of 11mm + 0.15mm for tolerance in material shrink
Connector gender / orientation	Motor shaft (male) into the pinion bore (female). The motor axis line lies parallel to the length of the frame, and the stage turns the drive through 90°.	Bevel gear must mesh at 90°
Rotor speed	Limited by RMS line to line voltage supplied by VFD. Rotor speed rated at 1360 rpm.	Specified at 60% to 90% of rated rotor speed. (816 rpm ↔ 1224 rpm)
Cable / harness type	Motor power cabling is covered by IF4.0.	-
Mounting arrangement	Motor fixed to the aluminium-profile gearbox frame on a profile bracket at the input end, with T-bolts.	Motor shaft aligned with the gearbox input within ± 2mm
Environmental limits	Outdoor operation on NWU Campus. The printed gears must withstand UV exposure and ambient temperature.	Temperature range 17°C to 36°C; enclosure or cover for the gearbox

Operating Sequence

1. The brake is clamped on the spool and the motor is stationary.
2. A hoist command is received. The VFD energizes the motor, and the brake electromagnet releases the spool.
3. The motor shaft drives the bevel pinion, which turns the drive through 90° into the herringbone stages.
4. The gearbox reduces the speed and multiplies the torque. The final herringbone flange drives the spool.
5. On a stop command, power loss or E-stop, the motor de-energizes and the spring-applied brake clamps the spool.

Interface Dependencies and Constraints

- The required motor torque and speed depend on the final gear ratio and spool diameter. The VFD characterisation in the Heavy Current Lab will confirm the motor's speed range for EEII 1 to configure drive electronics
- The motor must be accurately aligned with the gearbox input. Misalignment loads the printed bevel teeth unevenly and accelerates wear.
- The bevel input gear is still to be reprinted, and the bearings are on order. The interface can only be fit-tested once both are available.
- The motor shaft does not hold the load. Holding is done only by the spool brake, so the coupling only must carry drive torque, not static holding torque.

2.5.4 Interface IF12.0

Interface between FU 8 (System Controller) and FU 5 (Magnetic attachment)

Table 23: IF12.0 description

Field	Specification
Description / purpose	Uses the PWM and digital control signals from the System Controller (FU8) to drive the MOSFET switching module that energizes the three E-core electromagnets (FU5), and returns an analog coil current measurement to verify magnetic grip.
Connected components	System Controller (FU8) ↔ MOSFET Switching Module & Current Sensor ↔ Three-Point Electromagnet Attachment (FU5)
Interface owner(s)	Magnetic attachment: EEII 2; MCU control & telemetry: REII ; Mechanical coupling (IF11.0): MCTR 2
Reference drawing(s)	Figure 1 (System Block Diagram, REF: F/U 5.0)
Operating modes	Idle (24 V DC supply on, PWM = 0%, magnet de-energized); Energized / Holding (PWM at 8.33% duty cycle supplying ~2 V / 7.5 A total across three parallel coils); Off (24 V DC supply switched off).

Physical / Mechanical Interface

Table 24: IF12.0 Physical Description

	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	2.54 mm Dupont jumper wires from the MCU board to the logic-level converter / MOSFET switching module and current sensor; screw terminals with SWG 16 / SWG 19 copper wire to the electromagnet coils. Parameter	SWG 16 current rating: 3.7A SWG 19 current rating: 2.8 A
Connector gender / orientation	Female Dupont sockets on MCU board headers to male header pins on the switching and current-sensor modules.	[Reference]

Mounting arrangement	Core dimensions: 200 mm width × 100	MOSFET switching module and current
Environmental limits	[Temperature, vibration, moisture,	50°C-60°C

Electrical Interface

Table 25: IF12.0 Electrical Description

Parameter	Nominal requirement	Minimum / maximum / tolerance
DC Supply voltage	24V DC	Maximum
AC PWM voltage	3V– 5V AC	Minimum and maximum
Switching module current	40mA	Operation current
Electromagnet Current	7.5A - (2.5A per core in parallel)	Operational
Signal type	CMOS PWM	8.33% Duty Cycle
Logic levels	Low<3V, 3V≤High≤5V	5V max
Input / output impedance	3V=15mΩ, 5V=6mΩ	20mΩ max
Ground / reference	Grounded and isolated	-

Pinout and Signal Definitions

Table 26: IF12.0 Pin definitions

Pin / terminal	Signal name	Direction (relative to)	Electrical characteristics	Function / notes
J7.5 (Silkscreen 4 / IO4)	MAG_PWM (VIN)	Output (from MCU)	3.3 V PWM (8.33% duty)	PWM control signal from MCU (stepped to 5 V via logic level converter into MOSFET switch module)
J3.6 (EXP_PortA GPA3)	MAG_LOGIC	Output / Input	3.3 V Digital Logic	Auxiliary digital enable / status line for the electromagnet switching circuit
J3.12 (Bottom of J3)	GND	Reference	0 V Digital GND	Common digital ground reference between MCU board and MOSFET switching module
J6.8 (ADC_Port AN3)	MAG_CURR_IN	Analog Input (to MCU)	0 – 3.3 V Analog	Electromagnet coil current sensor input to on-board 24-bit ADC (monitors 7.5 A nominal coil current)
J6.9 (ADC_Port GND)	GNDREF	Reference	0 V Analog GND	Filtered low-noise analog ground return for the electromagnet current sensor

Switch Module Terminal 3	DC IN (SIG-01)	Power Input	24 V DC (from FU10)	24 V DC power input from secondary power supply (IF8.0) into MOSFET switching module
Switch Module Terminal 4	DC OUT (SIG-02)	Power Output	~2 V DC Scaled (7.5 A)	PWM-scaled DC output driving the three parallel E-core coils (2.5 A per core)

Data / Communication Interface

Table 27: IF12.0 Communication description

Parameter	Specification
Physical layer	3.3V
Frequency	Up to 20kHz

Operating Sequence

1. The DC supply will be turned on before operation of the electromagnet.
2. The controller will send a 3.3V PWM signal to the switching module at an 8.33% duty cycle.
3. The 3.3V input will be adjusted using a Logic Level Converter to 5V.
4. The Duty cycle reduces the DC input of 24V to approximately 2V.
5. This will be applied in parallel to the coils which will energize and generate around 2.5A
6. The applied current will generate an appropriate magnetic field.
7. When the PWM voltage drops below 3.3V the Switching module does not engage and the DC supply is disconnected
8. The Electromagnet Stops generating a magnetic field.
9. The electromagnet remains in an Idle state until the PWM signal rises.
10. When the DC supply is turned off the electromagnet cannot function and is in the OFF state

Interface Dependencies and Constraints

- The simulation was run in FEMM, This simulation was originally designed with a massive 1mm airgap, and repeated calculations revealed that the current would need to be much higher than necessary.
- In reality, the true air gap would be very small when the plates are attached. This led to the simulation being adjusted to 0.5mm. This discrepancy from reality means that the force would be much higher than the simulated value.
- The windings around the E-cores were designed at the maximum temperature the cores would experience; this means that the starting force would be higher than the operational force.

2.5.5 Interface IF10.0

This is the interface between the Remote Controller (FU6) and the System Controller (FU8).

Table 28: : IF10.0 description

Field	Specification
Description / purpose	Provides two-way wireless communication between the operator PC dashboard (FU6) and the crane's main controller board (FU8). Sends operator movement and magnet commands to the crane and returns live sensor readings and fault alerts back to the PC.
Connected components	PC Operator Dashboard ↔ USB Ground-Station Microcontroller ↔ Ground 2.4 GHz Radio Module ↔ [2.4 GHz Wireless Link IF10.0] ↔ Crane 2.4 GHz Radio Module ↔ Crane System Controller (FU8).
Interface owner	REII (R. Nel 40496325) — Ground station, wireless link, 32-byte data packet format, and PC dashboard; shared with MCTR 1, MCTR 2, EEII 1, and EEII 2 on FU8.
Reference drawing	Figure 1 (System Block Diagram, I/F10 and I/F-E1.0); Main Controller Board Wiring Diagram.
Operating modes	(1) USB Check Mode (verifies cable connection and radio settings); (2) Low-Speed Range Test Mode (Channel 80, 250 kbps for bench diagnostics); (3) Normal Crane Control Mode (Channel 77 / 2477 MHz, 2 Mbps, 32-byte command and automatic 32-byte sensor reply); (4) Safe Hold Mode (stops motors if wireless signal is lost).

Physical / Mechanical Interface

Table 29: IF10.0 Physical Description

Parameter	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	PC to Ground Station: standard USB cable. Radio modules to microcontrollers: 2.54 mm pitch pin headers (8-pin radio socket to 10-pin comms header and 12-pin digital header on the crane board). Antenna: screw-on 50 Ω RP-SMA antenna connector.	Standard USB 2.0 and 2.54 mm pin headers
Connector gender / orientation	Radio modules have a 2×4 male pin header (Pin 1 is the square ground pad) and a female antenna jack; the crane board uses female jumper leads to the keyed male board headers.	Verify 3.3 V and GND orientation before power-on (connecting 5 V will damage the 3.3 V radio)

Pin / contact count	USB cable: 4 pins. Each nRF24L01+ radio module: 8 pins total (7 connected: 3.3 V power, ground, chip enable, chip select, SPI clock, data in, and data out; interrupt pin left unused).	7 active wires per radio module
Cable / harness type	Shielded USB cable between PC and ground station; short 24–26 AWG jumper wires between the crane board and crane radio module.	USB cable ≤ 2.0 m; radio jumper wires ≤ 150 mm to keep signals clean
Mounting arrangement	Ground station sits next to the operator PC with its antenna vertical; crane radio mounts inside the enclosure with its antenna vertical and clear of the 220 V VFD power cables.	Use low radio power when testing closer than 2 m on the bench; full power at 5–10 m on the crane rail
Environmental limits	Indoor laboratory environment (0 °C to +45 °C).	Tested from 1 m to 190 m line-of-sight; normal crane operating distance is 5–10 m

Electrical Interface

Table 30: IF10.0 Electrical Description

Parameter	Nominal requirement	Minimum / maximum / tolerance
Supply voltage	Ground station: 5.0 V from PC USB (stepped down to 3.3 V on the dongle, drawing 0 mA from the crane). Crane radio: 3.3 V DC supplied from the crane controller board.	Radio supply: 3.0 V min / 3.3 V nominal / 3.6 V max
Current	Ground-station controller: ~40–80 mA. Radio module: ~115 mA short burst when transmitting at full power, ~24 mA when listening, < 1 mA on standby.	3.3 V supply must handle 150 mA peaks without dropping below 3.0 V
Signal type	USB serial data between PC and ground station; 4-wire SPI data bus plus 1 digital enable line between each microcontroller and its radio; 2.4 GHz wireless radio wave in air.	SPI clock speed capped at 4 MHz for clean signal edges
Logic levels	3.3 V digital logic on all radio and microcontroller pins (LOW = 0.0 V, HIGH = 3.3 V).	LOW: −0.3 to 0.8 V; HIGH: 2.0 to 3.6 V
Input / output impedance	Antenna port: 50 Ω matched impedance.	Antenna must be screwed on before turning power on
Ground / reference	Ground station uses the PC USB ground (completely separated from the crane's 220 V power system by the wireless link); crane radio shares common digital ground with the crane controller.	Full electrical separation between operator PC and crane power

Protection	A 10 µF electrolytic capacitor and a 100 nF ceramic capacitor are soldered directly across the 3.3 V and GND pins of both radio modules.	Prevents sudden voltage drops during 115 mA transmit bursts

Pinout and Signal Definitions

Table 31: IF10.0 Pin definitions

Radio Pin	Crane Main Board Pin	Ground-Station Pin	Signal Name & Direction	Function / Wiring Notes
nRF Pin 1	J1.5 & J1.6 (GND)	Dongle GND	GND (Reference)	0 V digital ground return; 10 µF + 100 nF capacitors soldered across Pins 1 & 2
nRF Pin 2	J1.8 (+3.3VP)	Dongle 3V3	VCC (3.3 V Power)	3.3 V DC power supply to radio (WARNING: never connect to J1.7 which is +5 V)
nRF Pin 3	J3.3 (EXP_PortA GPA0) (ref GND on J3.1)	Dongle GPIO 4	CE (Output from MCU)	Radio Chip Enable (3.3 V logic). Set HIGH on the crane at startup so the radio listens continuously
nRF Pin 4	J1.1 (SPI_xCS Square Pad)	Dongle GPIO 5	CSN (Output from MCU)	SPI Chip Select (active LOW). Selects the radio module on the shared SPI bus
nRF Pin 5	J1.3 (SPI_CLK)	Dongle GPIO 18	SCK (Output from MCU)	4 MHz SPI clock signal from microcontroller to radio
nRF Pin 6	J1.2 (SPI_MOSI)	Dongle GPIO 23	MOSI (Output from MCU)	SPI data line from microcontroller into radio (sends commands and 32-byte packets)
nRF Pin 7	J1.4 (SPI_MISO)	Dongle GPIO 19	MISO (Input to MCU)	SPI data line from radio into microcontroller (reads status and incoming 32-byte packets)
nRF Pin 8	Unconnected	Unconnected	IRQ (Unused)	Left unconnected on both sides to save pins; the microcontroller checks radio status over SPI

Data / Communication Interface

Table 32: IF10.0 Communication description

Parameter	Specification
Communication links	Wireless link (IF10.0): 2.4 GHz radio on Channel 77 (2477 MHz). Crane board bus: 4 MHz SPI + digital enable. PC to ground station (IF-E1.0): USB serial at 115 200 bit/s.
Data speed	Normal wireless speed: 2 Mbps (keeps radio time short to avoid collisions). Diagnostic test speed: 250 kbps on Channel 80. USB serial speed: 115 200 bit/s.

Wireless channel & Wi-Fi avoidance	Operates on Channel 77 (2477 MHz), which sits above campus Wi-Fi (Channels 1, 6, and 11) and measured 0% background interference. A 5-channel group plan [4, 50, 73, 77, 81] prevents interference between project groups.
Data format	Fixed 32-byte data packets in both directions using simple whole numbers (millimeters for position, mm/s for speed, tenths of a kg for weight, and tenths of an amp for current) so no decimal rounding errors occur.
Update rate & automatic reply	20–50 updates per second. Whenever the crane receives a 32-byte command packet, its radio automatically sends back the latest 32-byte sensor reading inside the confirmation reply.
Response time (latency)	A 32-byte command and 32-byte sensor reply take ~0.32 ms in the air, with up to 5 automatic retries (0.5 ms apart). Worst-case delivery time is under 4.15 ms (measured average is 0.89 ms).
Error checking & duplicate filtering	Every wireless packet includes a built-in 16-bit error check (CRC), a start marker byte (0xA5 for commands, 0x81 for telemetry), and a packet counter (0–255) so corrupted or duplicate packets are ignored.
Timeout / lost-link safety	If all 5 hardware retries fail (~4.15 ms), the dashboard is notified immediately. If the crane loses wireless contact, its onboard timer automatically stops the motors, applies the hoist brake, and keeps the magnet energized if holding a load.

Downlink Command Payload Map (PC Dashboard to Crane Controller FU8 - 32 Bytes)

Table 33: IF10.0 Downlink command payload map

Byte	Field Name	Type	Encoding / Scale	Function & Subsystem Target
0	magic	uint8 (1 byte)	0xA5 (Fixed ID)	Identifies a valid command packet; any packet not starting with 0xA5 is ignored.
1	msg_type	uint8 (1 byte)	1 = Ping Test, 2 = Move Command	Tells the crane whether this packet is a connection check or an active movement command.
2	sequence	uint8 (1 byte)	0 to 255 (counts up)	Packet counter used to ignore duplicate packets and echoed back by the crane.
3	mode_flags	uint8 (8 bits)	b0=Soft Stop, b1=Reset, b2=Auto, b3=Zero Pos, b4=Hold Brake	On/off control flags for software stop, clearing faults, automatic mode, zeroing position, and holding the brake.
4	trolley_cmd	int8 (signed)	−100% to +100%	Target horizontal carriage speed and direction for MCTR 1 (drives PWM on J7.9 and direction on J4.4–J4.5).
5	hoist_cmd	int8 (signed)	−100% to +100% (+Up / −Down)	Target vertical hoisting speed and direction for EEII 1 VFD (J7.6–J7.8, J3.4) and MCTR 2 Brake (J4.8).
6	magnet_state	uint8 (1 byte)	0 = OFF, 1 = ON	Turns the EEII 2 electromagnet ON or OFF (drives PWM on J7.5 and logic on J3.6).
7	segment_idx	uint8 (1 byte)	0 to 255	Step number for smooth anti-sway or automated path movement.

8..31	padding[24]	24 bytes	All 0x00	Unused spare bytes set to zero, so the packet always fills the required 32-byte radio size.

Uplink Telemetry ACK-Payload Map (Crane Controller FU8 to PC Dashboard - 32 Bytes)

Table 34: IF10.0 Uplink telemetry ACK-payload map

Byte	Field Name	Type	Encoding / Scale	Function & Subsystem Target
0	magic	uint8 (1 byte)	0x81 (Fixed ID)	Identifies a valid sensor reply packet from the crane.
1	seq_ack	uint8 (1 byte)	0 to 255	Repeats the command counter (Byte 2) so the dashboard knows which command triggered this reading.
2..3	trolley_pos	int16 (2 bytes)	Millimeters (1 mm per step)	Horizontal carriage position along the rail measured by the MCTR 1 wheel encoder (J7.12).
4..5	hoist_pos	int16 (2 bytes)	Millimeters (1 mm per step)	Estimated vertical hook height calculated from VFD frequency (EEII 1) and spool geometry (MCTR 2).
6..7	trolley_vel	int16 (2 bytes)	mm/s (1 mm/s per step)	Horizontal carriage speed calculated from the MCTR 1 wheel encoder pulse rate (J7.12).
8..9	hoist_vel	int16 (2 bytes)	mm/s (1 mm/s per step)	Estimated vertical lifting/lowering speed from VFD frequency (EEII 1) and spool radius (MCTR 2).
10..11	load_cell	uint16 (2 bytes)	kg × 10 (0.1 kg per step)	Hook load weight measured by the MCTR 2 load cell on J6.5 & J6.6 (e.g., 185 = 18.5 kg).
12..13	vfd_current	uint16 (2 bytes)	Amps × 10 (0.1 A per step)	Hoist drive current measured by the EEII 1 current sensor on J6.7 (e.g., 42 = 4.2 A).
14	magnet_current	uint8 (1 byte)	Amps × 10 (0.1 A, 0–25.5 A)	Electromagnet coil current measured by the EEII 2 current sensor on J6.8 (nominal 75 = 7.5 A).
15	magnet_status	uint8 (1 byte)	0=OFF, 1=ON (Holding), 2=Fault	Confirms whether the EEII 2 electromagnet is off, actively holding a load, or experiencing a current fault.
16	error_code	uint8 (1 byte)	0 = OK, 1..255 = Fault ID	Main fault number displayed as a clear warning message on the PC dashboard.
17	fault_flags	uint8 (8 bits)	1 bit per fault type	Shows active faults at once: b0=VFD Overcurrent, b1=Overload (>20 kg), b2=Brake Fault, b3=Rail Limit, b4=Magnet Fault, b5=Encoder Fault, b6=VFD Fault, b7=Wireless Timeout.
18	system_health	uint8 (8 bits)	1 bit per status indicator	Shows live crane status: b0=Wireless Link OK, b1=E-Stop Pressed, b2=Auto Mode, b3=Hoist Brake Clamped, b4=VFD Enabled.
19..31	padding[13]	13 bytes	All 0x00	Unused spare bytes set to zero to complete the fixed 32-byte packet.

Operating Sequence

- **Startup & Pin Setup:** The ground-station unit powers up from the PC USB port. On the crane, the main controller turns on the 3.3 V supply rail, sets all SPI select pins HIGH so no devices clash, starts the 4 MHz SPI bus on Header J1, and sets the radio enable pin (J3.3) HIGH so the crane radio listens continuously.
- **Connection Check:** The PC dashboard opens the USB serial link, checks that the ground-station radio is set to Channel 77 at 2 Mbps, and sends a test ping packet to confirm the crane responds.
- **Sending Commands:** As the operator moves controls on the dashboard, the PC calculates smooth acceleration steps (to prevent load swing), packs the 32-byte command packet with a new sequence number, and sends it over USB to the ground-station radio for wireless transmission.
- **Execution & Automatic Sensor Reply:** When the crane radio receives a valid command packet, it automatically sends back the pre-loaded 32-byte sensor packet inside the confirmation reply. The crane controller then updates the motor PWM, direction, brake, and magnet pins and loads the latest sensor readings for the next reply.
- **Lost-Link & Emergency Stop Safety:** If wireless packets stop arriving or a sensor detects an unsafe condition (such as a load over 20 kg), the crane controller stops horizontal and vertical motion and clamps the hoist brake while keeping the electromagnet powered if a load is hanging. The physical red emergency-stop button cuts high-voltage power directly regardless of software.

Interface Dependencies and Constraints

- Both nRF24L01+ radio modules require a 10 µF electrolytic and 100 nF ceramic capacitors soldered across their 3.3 V and GND pins to prevent voltage drops during transmit bursts.
- The SPI clock on Header J1 must be kept at or below 4 MHz for reliable communication, and only one SPI device (radio, ADC, or SD card) may have its chip-select line pulled LOW at a time.

2.5.6 Interface IF13.0 & IF15.0

Interface between FU 8 (System Controller), FU 1 (Load Cell — IF13.0), and FU 2 (Hoisting Mechanism Fail-Safe Brake — IF15.0).

Table 35: IF13.0 & IF15.0 description

Field	Specification
Description / purpose	Reads the on-hook tension load from the S-type load cell (IF13.0) to monitor payload weight and prevent overloads and switches the hoisting gearbox's fail-safe electromagnetic brake (IF15.0) to release during lifting/lowering and clamp automatically when stopped or if power fails
Connected components	IF13.0: S-Type Load Cell (FU1) ↔ System Controller (FU8) IF15.0: System Controller (FU8) ↔ Hoist Brake MOSFET Switch Module (FU2)
Interface owner(s)	MCU side – REII Hoist Brake & Load Cell side – MCTR 2

Operating modes	Brake Engaged / Safe Hold (default 0 V output holds load stationary); Active Hoisting (brake energized HIGH to release drum while VFD runs); Overload Inhibit (lifting blocked if load exceeds 20 kg limit).

Physical / Mechanical Interface

Table 36: IF13.0 & IF15.0 Physical Description

Parameter	Requirement / specification	Tolerance / limits / reference
Connector / coupling type	Standard 2.54 mm pitch Dupont jumper connectors from the load cell cable and LR7843 brake MOSFET driver module to the MCU carrier board.	Standard 2.54 mm pitch
Connector gender / orientation	Female Dupont sockets on MCU headers to male pins on the brake module and load cell lead adapter.	Observe + / – signal polarity on the 4-wire load cell bridge
Pin / contact count	Load cell (IF13.0): 4 wires (Excitation +, Excitation −, Signal +, Signal −). Hoist brake (IF15.0): 2 control wires (Brake ON/OFF signal and Digital GND).	6 connections total on main board
Cable / harness type	Shielded 4-core cable for the load cell (routed down the hoist path with slack for vertical travel); 24–26 AWG stranded wire for the brake control signal.	Keep load cell cable away from 220 V VFD motor leads
Mounting arrangement	S-type load cell suspended in tension between the hoist wire rope (IF14.0) and magnetic attachment (IF11.0); brake module mounted on the hoist gearbox frame (FU2).	Designed for 20 kg max payload (35 kg mechanical safety rating)
Environmental limits	Indoor laboratory crane environment (0 °C to +45 °C).	-

Electrical Interface

Table 37: IF13.0 & IF15.0 Electrical Description

Parameter	Nominal requirement	Minimum / maximum / tolerance
Supply voltage	Load cell excitation: 3.3 V DC low-noise analog rail from MCU board. Brake coil power: external DC supply switched via LR7843 MOSFET module.	3.3 V ± 0.15 V excitation
Current	Load cell bridge: ~10 mA excitation draw. Brake control signal: < 5 mA logic gate drive.	Max 20 mA on digital logic pin

Signal type	IF13.0: Millivolt differential analog bridge signal (or 2-wire digital serial if external HX711 amplifier is used). IF15.0: 3.3 V digital HIGH/LOW ON/OFF signal.	Differential mV signal into 24-bit ADC; 3.3 V CMOS brake logic
Logic levels	Brake control (IF15.0): HIGH (3.3 V) = energize brake coil to release spool; LOW (0.0 V) = de-energize coil so spring clamps the brake pad (fail-safe hold).	LOW: 0.0 – 0.8 V; HIGH: 2.0 – 3.3 V
Input / output impedance	Filtered analog ground reference for the load cell bridge; common digital ground for the brake MOSFET trigger module.	Separate analog and digital ground returns on carrier board
Ground / reference	On-board 1 kΩ + 2.2 nF low-pass filter on ADC inputs; spring-applied brake engages automatically on power loss.	Software interlock trips if load > 20.0 kg
Protection	IF13.0: Millivolt differential analog bridge signal (or 2-wire digital serial if external HX711 amplifier is used). IF15.0: 3.3 V digital HIGH/LOW ON/OFF signal.	Differential mV signal into 24-bit ADC; 3.3 V CMOS brake logic

Pinout and Signal Definitions

Table 38: IF13.0 & IF15.0 Pinout Description

Main Board Pin	Signal Name	Direction (rel. to MCU)	Electrical Spec	Function / Wiring
J6.1 (Top Square Pad)	+3.3VP (EXC+)	Output (Power Rail)	3.3 V DC Analog Rail	IF13.0: Positive excitation supply (Red wire) to S-type load cell bridge
J6.2 (ADC_Port GND)	GNDREF (EXC−)	Reference	0 V Analog GND	IF13.0: Negative excitation / ground return (Black wire) for load cell bridge
J6.5 (ADC_Port AN0)	LOADCELL_P (SIG+)	Analog Input	Differential mV (+)	IF13.0: Load cell positive tension signal (Green/White wire) to on-board 24-bit ADC AIN0
J6.6 (ADC_Port AN1)	LOADCELL_N (SIG−)	Analog Input	Differential mV (−)	IF13.0: Load cell negative tension signal (White/Green wire) to on-board 24-bit ADC AIN1

J3.7 & J3.8 (Optional)	HX711_DOUT / SCK	Digital In / Out	3.3 V Digital Logic	IF13.0 Optional Alternative: If an external HX711 board is used instead of Header J6, wire DOUT to J3.7, SCK to J3.8, +3.3VP to J3.11, and GND to J3.1
J4.8 (EXP_PortB GPB5)	HOIST_BRAKE	Output	3.3 V Digital Logic	IF15.0: Fail-safe hoist brake ON/OFF signal to LR7843 MOSFET switch (HIGH = Release brake, LOW = Engage brake)
J4.12 (Bottom of J4)	GND	Reference	0 V Digital GND	IF15.0: Common digital ground return for the hoist brake MOSFET module

Data / Communication Interface

Table 39: IF13.0 & IF15.0 Communication Description

Parameter	Specification
Load cell sampling (IF13.0)	24-bit ADC conversion at 20–50 Hz; calibrated in software into tenths of a kilogram (0.1 kg steps, e.g., 200 = 20.0 kg) for wireless telemetry.
Brake actuation (IF15.0)	Direct digital HIGH/LOW state synchronized with hoist VFD commands; brake releases 50ms before motor start and clamps immediately when hoist speed reaches zero or a fault/link drop occurs.

Operating Sequence

- **Startup & Tare:** System Controller powers the load cell on Header J6, keeps the hoist brake engaged (LOW on J4.8), and records the zero-load tare offset.
- **Lift / Lower Command:** When a vertical hoist command arrives, the controller checks that measured load is within the 20 kg limit, starts VFD excitation, and drives J4.8 HIGH to release the spring brake.
- **Continuous Load Monitoring:** During lifting, the controller reads the load cell continuously and sends the weight in 0.1 kg increments to the operator dashboard.
- **Stop / Fail-Safe Hold:** When the operator stops hoisting or if a wireless timeout or overload occurs, J4.8 is driven LOW immediately so the spring brake clamps the spool mechanically.

Interface Dependencies and Constraints

The hoist brake must default to the clamped (safe) state whenever the microcontroller is unpowered or resetting. The load cell must be calibrated with known test masses prior to final rail integration so the 20 kg overload interlock triggers accurately.

3 Main Design Decisions

3.1 MCTR 1 – Lateral

The lateral motion subsystem has three main functions. Firstly, to be able to roll along the given track whilst under load, which is done by the carriage system. Secondly, to drive along the track using DC motors. Lastly, to keep track of the position and or speed using motor encoders.

3.1.1 Carriage system

After initial testing of Milestone 2, the first prototype of the carriage system was revised to prototype version two. In this revision the following decisions were made in an attempt to solve the respective problems identified.

Table 40: Carriage system design decisions

Problem Identified	Design Choice	Reason	Verified (Y/N)
Wheel axles were not properly supported and caused sagging under load.	Use a wheel bearing bracket with 2 bearings, separated at a given distance along the axle.	Bearing ID's are close to axle OD's. Each bearing restricts perpendicular axle movement, with greater bearing separation causing greater restriction.	Y
Wheels jammed on track welds.	Use at least 60mm diameter wheels.	Larger diameter wheels roll over welds easier.	Y
Track left and right side flanges are not equally thick and track has large welds.	Do not use bottom driven wheels. Rather use bottom support wheels that do not engage with the track unless the carriage starts tipping over.	The welds still cause collision with larger wheels, and when hanging a load will cause the carriage to tip and lift front wheels off the track. This risks the carriage falling off. Support wheels can possibly prevent this.	N
Wheel spacing adjustment screw introduces additional degrees of freedom (DOF).	Remove wheel spacing adjustment screw, experimentally determine a good wheel spacing and space them at assembly.	This removes the additional DOF making the structure more rigid and load bearing capacity higher.	Y
Long side plates causes a lever effect, amplifying the twisting motion of the plates, causing sagging under load.	Use aluminium extrusions and plastic brackets to build a rigid frame.	This mostly eliminates the additional sagging due to the lever effect.	Y
Tight wheel spacing restricts motion around track bends.	Use wheel spacing that only makes contact when turning, to reduce restrictions and friction.	Wheel side flange contact is only needed when taking bends, and only on one side at a time.	Y

3.1.2 Motor drive system

The carriage needs to be driven by DC motors. From a combination of budget, power and GPIO pin constraints, it was chosen to use only 2 DC motors, one on either side. This is the minimum amount of motors required to produce the torque required for acceleration.

The next decision was whether to use only two driving wheels or drive all four wheels using a centre pinion gear attached to the gearbox. The latter was selected in an attempt to increase the amount of possible contact points and thus increase the power delivered by the carriage to the rail.

The current motors require a speed increase and torque reduction, therefore an additional 1.5:1 gearbox is needed which is made from 3D printed gears.

A single motor driver can be used to drive two motors and both always turn in the same direction, at the same speed. Thus, the two motors will be driven from a single motor output of the controller, to halve the number of pins needed from the MCU.

3.1.3 Encoders

The current motors have built-in quadrature encoders that will be used to provide position and speed feedback to the controller. As both wheels are turning together, only one encoder will interface with the MCU. Additionally, because the direction will be known from within the MCU code, a single encoder channel can be used.

3.2 MCTR 2- Spooling

The main design decisions below cover the design load, gearbox architecture, spool and cable, fail-safe brake, and on-hook load measurement. Each decision is linked to the system requirements it addresses and to the work packages it affects.

3.2.1 Design load and safety margin

The system specification requires the hoist to lift a payload of up to 20 kg (FR4). The gearbox and spool are instead designed for a load of 30–35 kg. This margin covers the weight of the magnetic attachment and the load cell, dynamic loading during acceleration and braking, and variability in the payload. At 35 kg on the 70 mm drum, the required drum torque is:

$$T = m \cdot g \cdot r = 35 \cdot 9.81 \cdot 0.035 = 12\ N \cdot m$$

This design load sets the torque required from the EEII I motor, the gear tooth sizing, and the measurement range of the load cell.

3.2.2 Gear type and gearbox architecture

Herringbone gears were selected over spur gears. They offer a significantly higher load capacity, smoother and quieter meshing, and self-cancelling axial thrust, which removes the need for thrust

bearings on the gear shafts. A gear module of 3 was chosen for the tooth size. Since a worm gear configuration is excluded (DR2), speed reduction is achieved with a bevel gear input stage followed by the combination of herringbone gear stages. The bevel stage turns the drive through 90°, so the motor can lie parallel to the length of the frame and keep the assembly compact. The gearbox frame is a rectangular structure of 20 × 20 mm aluminium square profile, joined with die-cast corner brackets.

3.2.3 Spool and cable design

A 3 mm stainless steel wire rope is proposed as the main hoist cable. The spool drum is 70 mm in diameter, with a helical groove printed into its surface so the cable winds evenly onto the drum. The spool has two flanges, and one of them is formed as a herringbone gear, so the final gear stage drives the spool directly.

3.2.4 Brake conception

Because the worm gear configuration is excluded (DR2), the hoist cannot rely on a self-locking gearbox to hold the load, so an active brake is required. An electromagnetic, spring-applied brake was chosen over the alternatives because it is fail-safe by default and therefore satisfies FR6. When the motor is running, the electromagnet is energised and pulls the brake pad clear of the spool. When power is lost or the motor is idle, the electromagnet de-energises and a spring presses the brake pad against the spool flange. This holds the suspended load and stops the cable from unwinding.

3.2.5 Load measurements

An S-type load cell is mounted between the hoist cable and the magnetic attachment, where the suspended load places it in tension. The load cell outputs a millivolt-level signal. An HX711 amplifier is selected to amplify and digitise this signal, and it transmits the reading to the controller over two digital GPIO lines. This avoids using the controller's on-board ADC, which lacks the resolution for a signal this small.

3.3 EEII 1 – Drive circuit

The VFD drive circuit use a rectifier to convert the provided single phase AC source voltage to a DC bus voltage. The DC bus voltage is converted to a sinusoidal shape PWM signal using a pair of power switches(IGBT's) to form the high and low side of the sinusoidal wave. The IGBT's are controlled by a driver IC to enable the sequence in which the high and low side switching pair must operate. The driver IC is controlled by a PWM signal from the MCU, which is isolated from the high voltage circuit with an opto- coupler. This configuration is employed three times to provide a three-phase supply to the induction motor.

3.3.1 Rectifier

A full wave bridge rectifier was chosen to convert the single-phase AC supply voltage to a DC bus voltage. The GBJ1510 rectifier provided a forward AC current limit of 15A with a DC voltage rating of 1000V. For the single-phase AC 230V input, the approximate DC bus voltage output was determined as 325V.

3.3.2 Driver IC's

To ensure the protection of the IGBT's against shoot through fault current, the high and low gate control outputs can never be at a high logic level simultaneously. To prevent this, the IR2104 driver IC was chosen since it provides built in deadtime as a safety precaution. Given the approximate DC bus voltage, the driver IC's are rated at 600V.

3.3.3 IGBT's

The induction motor is rated at 180W,and from the simulation the three phase line to line voltage was estimated at 110V between phases of the motor. The expected running current per phase was calculated as 1.63A. Since the starting current of an induction motor is usually much higher, the maximum circuit current rating was chosen as 12A. The FGH75T65SHDT IGBT's from onsemi was chosen with a current rating of 75A and a voltage rating of 650V. This current rating only applies to specific conditions with extreme heat sinks. More affordable heatsinks were chosen which should keep the junction temperature of the devices in check. The IGBT's also provide suitable rise and fall times for switching. Due to the inductive load of the motor, flyback diodes were required to create a return path for the current. The diodes introduce a ripple on the DC bus voltage which is why a capacitor was required over the DC bus.

3.3.4 Opto-Couplers

In the event of a fault current, the MCU is exposed to damage. The 4N25X opto-couplers were chosen to provide optical isolation between the MCU and the drive circuit. To achieve isolation, the diode side of the opto-coupler made use of the MCU ground, while the BJT side of the opto-coupler was connected to the ground of the drive circuit. The opto-couplers provided suitable rise time to manage the delay on the PWM signals.

3.4 EEII 2 – Magnet

The choices made in the design process as well as the reasoning behind these choices are provided in this section.

3.4.1 Core type

The design of the magnetic circuit started with the construction of the core types. As this can be chosen and the magnetic circuit relies heavily on the permeability of the chosen core, two options were investigated.

Table 41: Magnet design choices

Core type	Advantages	Disadvantages
C/U - type	• Light and easily manufacturable • 2 air gaps • Pole pair available	• Flux passes through one path • Higher current required due to saturation • Higher operational temperature
E - type	• 2 flux paths • Lower current required • 3 poles per core	• Heavier • Physical size constraints • 3 Airgaps

From these two both were designed to exceed 65N per core, as this is the weight requirement of the payload. The E-core showed more flux for less current while remaining in the linear region of the B-H curve. This is due to two paths available for the flux; this means that for the same flux density in the payload, the force can be up to twice as much. The E-core then surpasses the C-type core in terms of magnetic and electrical properties. The C-core is easier to construct and much lighter, but for a safer current level, the E-type core was chosen for the magnetic core.

3.4.2 Core dimensions

The dimensions of the core determine the total reluctance. Keeping this in mind, the pole areas were calculated to be 1600with this the core dimensions were decided as follows.

Table 42: Core dimensions summary

Component	Unit	Reasoning
A-core		This core area provides a big enough surface to limit the reluctance of the core
A-legs		The leg area is half of the core as the magnetic flux divides between the two legs
h		The height of the core was chosen to be small enough so that the cores don't obstruct the hoisting mechanisms and to limit the reluctance

w		The width of the core was chosen for symmetrical distribution over the payload.

With these specifications the reluctance and permeability of the core can be calculated and the required current and windings.

3.4.3 Current and conductors

The force of the electromagnet relies directly on the number of turns and the current through the windings (F=NI). So an increase in current means the number of turns can be less; however, the length of the wire used to set up the coil and the number of turns have severe financial implications. Having more current in the windings means that the conductor size needs to be bigger. If the current is lower, more windings are needed to achieve the same force. After investigation, it was found that thicker wire would be more expensive than adding more turns.

3.4.4 Switching module

After deciding that more turns will be used the current per core was found to be 2.5A. The three cores will be in parallel which results in 7.5A being pulled from the switching circuit to the cores. His means That an appropriate switching module was needed. The chosen module was a high-Power MOS trigger switch drive module PWM adjustment control board. The module is rated for 5-36V at 400W with a maximum current of 15A. This will be sufficient for the 7.5A at the 24V input which results in at most 180W if the output is considered it is 2V for 7.5A which is 15W. Both power values are well below the rated 400W and the module can handle if a fault occurs.

3.5 REII – Remote Control Dashboard and RF Ground Station

The remote-control and wireless subsystem was designed around four practical project requirements: (1) using the required 2.4 GHz nRF24L01+ radio modules, (2) fitting all commands and sensor readings into the radio's 32-byte packet limit, (3) sharing the crane microcontroller board cleanly with the other four subsystems without running out of pins, and (4) ensuring the crane stops safely on its own if the wireless connection ever drops. The main design choices and reasons are summarized below.

3.5.1 Ground-Station Microcontroller Selection and Power Architecture

A NodeMCU-32S (ESP32) board was chosen for the ground-station radio unit instead of an Arduino Nano. During early tests, the radio module drew short 115 mA current spikes when transmitting, which caused the smaller 3.3 V regulator on an Arduino Nano to drop out and reset. The NodeMCU-32S has a stronger 3.3 V regulator that handles these current bursts reliably and uses the same programming tools as the crane's main ESP32-S3 controller. Powering the ground station directly from the PC's USB port also keeps the operator PC electrically separate from the crane's 220 V motor drive power.

3.5.2 Wireless Channel Selection (Avoiding Campus Wi-Fi)

Standard campus Wi-Fi networks operate in the lower and middle parts of the 2.4 GHz band and caused up to 22.3% lost packets during initial lab tests. By scanning all 126 available radio channels, Channel 77 (2477 MHz) was found to sit cleanly above campus Wi-Fi with zero background interference. Running the radios at their fastest 2 Mbps data rate also cuts the time each packet spends in the air in half compared to 1 Mbps, resulting in 0% packet loss across the 1–10 m crane operating area. A 5-channel group plan (Channels 4, 50, 73, 77, and 81) was also set up so multiple student groups can run their cranes at the same time without interfering.

3.5.3 Automatic Sensor Replies and Fast Retry Limit

Instead of switching the crane radio back and forth between receive and transmit modes in software, the radio's built-in automatic reply feature (Auto-ACK) is used. Whenever the crane receives a 32-byte command packet from the PC, the radio hardware automatically attaches the latest 32-byte sensor reading to the confirmation reply. Automatic retries are limited to 5 attempts (0.5 ms apart) so a command either succeeds within 4.15 ms or is dropped immediately. We intentionally avoided queuing old failed commands in software so that delayed motor movements can never suddenly replay after a brief signal drop.

3.5.4 Wire Length Limits and Radio Power Decoupling

During initial bench testing, high clock speeds on the jumper wires caused signal noise, and radio transmit bursts caused small voltage dips on the 3.3 V power pins. Limiting the SPI communication clock between the microcontroller and the radio to 4 MHz cleaned up the digital signals while still transferring packets in a fraction of a millisecond. Soldering a 10 µF electrolytic capacitor and a 100 nF ceramic capacitor directly across the radio's 3.3 V and ground pins smoothed out the current spikes and eliminated power resets completely.

3.5.5 Main Board Pin Allocation Across All Five Subsystems

The shared crane controller board has a single high-speed pin header (Header J7) alongside two digital expander headers (Headers J3 and J4) and a dedicated communications header (Header J1). To prevent pin conflicts between group members, Header J7 is reserved strictly for signals that require fast hardware timers: EEII 1's three 20 kHz VFD PWM signals (J7.6–J7.8), EEII 2's electromagnet PWM signal (J7.5), and MCTR 1's motor speed PWM (J7.9) and wheel encoder pulses (J7.12). All simple ON/OFF digital signals across the group—including MCTR 1 motor direction (J4.4–J4.5), MCTR 2 hoist brake control (J4.8), EEII 1 VFD enable and fault lines (J3.4–J3.5), EEII 2 magnet enable (J3.6), and the REII radio enable pin (J3.3)—are placed on Headers J3 and J4. Meanwhile, the crane radio connects its data and power wires directly to the 10-pin communications header (Header J1), and all analog sensors (MCTR 2

load cell and EEII 1/EEII 2 current sensors) connect to the 24-bit analog header (Header J6). This gives every subsystem dedicated plug-in pins with zero soldering onto surface-mount board parts.

3.5.6 Simple Whole-Number Data Packets and Smooth Anti-Sway Control

- **Fixed 32-Byte Packets with Whole Numbers:** All commands and sensor readings are packed into fixed 32-byte packets using whole numbers—such as position in millimeters, speed in mm/s, hook weight in tenths of a kilogram (kg × 10), and current in tenths of an amp (A × 10). Using whole numbers avoids decimal formatting issues and fits every subsystem's data into a single radio packet.
- **Clear Fault Reporting:** Each return packet includes both a main fault number and an 8-bit fault checklist so the PC dashboard can show multiple warnings at the same time (such as an overload and a VFD trip).
- **Smooth Anti-Sway Speed Ramps:** Trying to stabilize a swinging load over a wireless link using fast back-and-forth corrections is unsafe if the wireless signal lags. Instead, the PC dashboard calculates smooth acceleration and deceleration ramps based on the rope length, so the carriage starts and stops without making the load swing. Each movement command is self-ending, so if the wireless link drops mid-move, the crane automatically brings the carriage and hoist to a safe stop.
- **Backup Software Stop vs. Physical Emergency Stop:** The stop button on the PC dashboard sends a software stop command over the wireless link, while the physical red emergency-stop switch cuts 220 V power directly in hardware as the primary safety cut-off.

4 Integration

Integration happens when all subsystems come together to form the whole. The success of integration depends on the compatibility of interfaces described in this document. In order to increase the likelihood of success, integration can first be tested before being finalised.

4.1 Pre-integration Test Plan

Before, attempting integration the interfacing subsystems need to be tested and their interfaces tested. This will show whether or not the interfaces need reworking or not.

4.1.1 REII – Remote Control Dashboard and RF Ground Station

- **Power & Wiring Check:** Use a multimeter before turning power on to verify that the crane radio receiver 3.3 V on pin J1.8 , that the 10 µF and 100 nF filter capacitors are connected across 3.3 V and GND, and that Both 50 Ω antennas are screwed on tightly.
- **USB & Radio Settings Check:** Connect both the ground-station microcontroller and the crane ESP32-S3 board to the PC via USB and run the built-in status check (ECHO & INFO) to confirm both radios respond on Channel 77 (2477 MHz) at 2 Mbps.
- **200-Packet Wireless Range & Speed Test:** Send 200 back-to-back test packets over the air between the ground station and crane board to confirm less than 5% packet loss and a round-trip response time under 4.15 ms.

- **32-Byte Command & Sensor Reply Test:** Send test command packets from the PC dashboard to the crane board and verify that all motor, brake, and magnet commands unpack accurately and that simulated sensor values display properly on the dashboard.

4.1.2 MCTR 1 – Lateral Movement

The Lateral movement pre-integration test plan is two-fold. First, the self-test is performed to check whether the subsystem is ready for integration. Second, the integration aspect is tested with REII and MCTR2. Integration with EEII1 does not need testing as of yet.

Self-test:

- After the carriage redesign it needs to be assembled and tested under load (without drive-train sub-system).
- The motor drive-train sub-system needs to be assembled and fit tested.
- A basic microcontroller sketch is then used to control the motor driver to test the motors without load.
- The system should then be placed onto the rail and loaded while the motor drive-train sub-system is attached.
- Thereafter, the motors are tested while loaded using a tethered arrangement to give motor commands via the serial interface from a laptop.

REII:

- The basic motor control should be tested with the given MCU, checking motor directions and PWM control.
- Thereafter, encoder readings can be tested before finalising the control and feedback implementation.

MCTR2:

- The mounting interface should be attached and tested to get an idea of the space available and possible placing for components.

4.1.3 MCTR 2 – Spooling

The spooling and gearbox system acts as the structural backbone of the integrated hoist, because the other subsystems either mount onto it or connect through it. Before can begin, the gearbox must be built with a frame and gear strong enough for the loads they will carry. The following must be completed before testing:

- All required gears printed according to the recommendations of the Increment 2 report

- Gearbox frame assembled
- All gears and the motor mounted in the gearbox frame

Onwe these are complete, the system can be tested. The motor will be connected to a VFD in the Heavy Current Lab to characterise it and to measure how the gearbox and spool perform across a range of drive frequencies.

MCTR 2 and MCTR 1

- The gearbox system will be fixed to the lateral motion system through dedicated mounting interfaces. Measurements will be taken from both assemblies to design the brackets or other connecting parts to join them.

MCTR 2 and EEII 1

- Once the motor has been characterised, the results from the VFD tests, such as the relationship between the drive frequency and output speed, will be given to EEII 1. They will use these values to configure the drive circuit, so it delivers the required motor RPM.

MCTR 2 and EEII 2

- The magnetic attachment will hang from the hoist cable, with the load cell as the physical connection point between them. Eye bolts fitted to both ends of the load cell let it carry the load tension and provide the anchor point for the magnetic attachment.

MCTR 2 and REII:

- Connect the S-type load cell to Header J6 (pins J6.1, J6.2, J6.5, J6.6) and the hoist brake MOSFET module to Header J4 (pins J4.8 and J4.12); apply known test weights to verify that the laptop dashboard displays the correct hook load in 0.1 kg steps and that the brake releases and clamps reliably when commanded.

4.1.4 EEII 1 – Electronic Drive circuit

A single-phase version of the drive circuit was tested and a DC bus voltage of about 15V was successfully switched (refer to progress section below). The following test plan was revised for the entire circuit.

Individual Testing (Low Voltage):

- With the PCB manufactured and delivered, the opto-couplers will be soldered onto the PCB with their corresponding resistors. A PWM signal will be applied to test the function of the opto-couplers.

- The driver IC's will be soldered onto the PCB with their corresponding resistors and capacitors and tested by application of the same PWM signal to the opto-couplers in order to measure the high and low side gate control signals.
- The IGBTS and corresponding diodes will be soldered onto the PCB. The PWM will be applied to the opto-couplers again to attempt the switching of a 15V DC bus voltage.

Individual Testing (High Voltage):

- The DC bus voltage will be increased in increments of 30V , up to 120V and the line-to-line RMS voltages will be measured.
- The rectifier will be connected to supply the bus voltage and the line-to-line RMS voltages will be measured.

REII and EEII 1:

- The MCU PWM signals will be applied to the opto-couplers to attempt the switching of a 15V DC bus voltage.
- The MCU PWM signals will be applied to the opto-couplers to attempt the switching of a 120V DC bus voltage.
- The induction motor will be connected to the VFD by a Delta connection with an attempt to start the motor.

MCTR 2 and EEII 1 and REII:

- The induction motor will be attached to the spooling mechanism in an attempt to test the spooling of a weight.

4.1.5 EEII 2 – Magnet Load attachment

- The communication from REII will be tested with no load to verify the operation of the switching module.
- The Electromagnet will then be tested using the PWM input from the microcontroller. The aim of this is to measure the current at the input to the coils. This will verify the buck converter and the switching module operation to the logic-level input.

4.2 Overall performance test plan

After integration, the success of the system will be tested with the performance requirements given. To do so, the performance of each subsystem has to be tested against the given requirements. The following lays out how that is planned to be done.

4.2.1 Subsystem 1 – MCTR1 (Lateral motion)

Performance target 1 states that the lateral motion subsystem should achieve 1 m/s (or greater) within 1 m of the departure point when no plates are attached. In order to test the performance of the system against this, the following test plan will be followed:

- The carriage will be placed at a selected starting point.
- The end point will be marked at a distance of 1 m from the starting point (straight section).
- The system will then be commanded to move forward until the end point is passed.
- The time from start to end point will be measured using a stopwatch.
- The speed is then calculated as 1 m divided by the time captured.

4.2.2 Subsystem 2 – MCTR2 (Spooling)

Performance target 2 states that the spooling subsystem should complete a 1.6m hoist stroke in 20s or more while carrying a payload of up to 20kg. This corresponds to an average lift speed of about 80mm\s:

$$V = 1.6m / 20s = 0.08mm/s$$

The following plan will be used to assess the system against this target:

- The magnetic attachment, carrying a 20kg test payload, will be lowered until the payload rests at a marked starting height.
- An end point will be marked 1.6m above the starting height.
- The system will be commanded to hoist until the payload passes the end point
- The time from start to end point will be measured with a stopwatch.
- The average lift speed is calculated as 1.6m divided by the measured time.

The fail-safe brake will be verified in the same test session:

- With the 20kg payload suspend at mid-stroke, the drive power will be removed.
- The brake should engage automatically and hold the payload.
- Any downward drift of the payload will be measured against a reference mark after 60s.

4.2.3 Subsystem 3 – EEII1 (Driver Electronics)

The performance targets for the VFD will be validated as follows.

Integration with Spooling :

- The E-brake will be applied after which the MCU reaches a state of emergency.
- The MCU apply an error signal to the control switch off the VFD.

Integration with MCU:

- The emergency error signal will be applied by the MCU after which it is expected that the High voltage side of the VFD is completely off/safe.
- The DC bus voltage will be measured with a multimeter.

Overall:

- The operation of the VFD will be tested by inspection of the operation of the induction motor.
- The overcurrent protection cannot be tested safely, however it can be identified in the circuit design.

4.2.4 Subsystem 4 – EEII2 (Electromagnetic attachment)

Integration for the electromagnetic attachment will rely on the hoist (Subsystem 2) load cell attachment and the microcontroller (Subsystem 5). To start subsystem 5 will be discussed

- Communication from the PWM signal to the switching module will be tested in no load conditions
- Communication from the current sensor to the microcontroller will be lab tested under light/ideal conditions.

The hoist integration:

- Collaboration in casing design for magnets allows the hoist designer to understand the possible connection possibilities.

4.2.5 Subsystem 5 - REII (Remote Control Dashboard and RF Ground Station)

Integration of the REII subsystem with the shared ESP32-S3 System Controller (FU8) and the other four subsystems (MCTR 1, MCTR 2, EEII 1, and EEII 2) is carried out step by step—starting from the PC USB link, then the 2.4 GHz wireless link, and finally the main board control pins; so each connection is verified before turning on high-voltage motor power to benchmark the performance:

Table 43: Overall test plan and performance verification

Integration Stage	Connected Elements	Performance Test & Verification Approach
1. PC to Ground Station (IF-E1.0 USB Link)	PC Dashboard ↔ Ground-Station Microcontroller (115 200 bit/s USB)	Test dashboard buttons, displays, and 32-byte packet formatting over the USB cable using simulated sensor values before turning on the wireless radios.
2. Wireless Radio Link (IF10.0 Air Link)	Ground nRF24L01+ Radio ↔ Crane nRF24L01+ Radio (Channel 77, 2477 MHz, 2 Mbps)	Run a 200-packet wireless test at low power on the workbench (< 2 m apart) so the close antennas do not overload each other, then switch to full power at the 5–10 m crane rail distance.
3. Crane Radio to Main Board (Headers J1 & J3.3)	Crane nRF24L01+ Radio ↔ ESP32-S3 System Controller (FU8)	Turn on the 3.3 V rail on J1.8, verify clean 4 MHz SPI communication on J1.1–J1.4, and set the radio enable pin (J3.3) HIGH so the crane receives commands continuously without interfering with the PWM pins on Header J7.

4. Full Group Signal Handshake on FU8	REII 32-Byte Packets ↔ MCTR 1, MCTR 2, EEII 1 & EEII 2 Pins	Verify with motors unpowered that dashboard commands drive the exact main board pins for MCTR 1 (J7.9, J4.4–J4.5), EEII 1 (J7.6–J7.8, J3.4), MCTR 2 brake (J4.8), and EEII 2 magnet (J7.5, J3.6), and that live readings from the encoder (J7.12), load cell (J6.5–J6.6), and current sensors (J6.7, J6.8) appear accurately on the PC.

5 Progress

5.1 MCTR 1 – Lateral Motion

At the time of writing the report, the carriage and required motor control subsystem have reached design iteration 2. The first motor tests were performed using a simple microcontroller sketch to control the motors via the serial terminal, while under 15kg of load (approximately half of the maximum load). This showed promising results in addition to design flaws that need to be revised before conducting further tests.

Promising results:

- The design idea of the motor gear train and geared wheels (all-wheel drive)
- Motor mount concept and gearbox mounting concept
- Carriage frame and brackets design

Problems identified:

- The gear mounting on the main axle was not sufficient and caused gear slipping
- Many gearbox enclosure dimensions were incorrect which caused mounting and assembly issues

Implementation evidence:

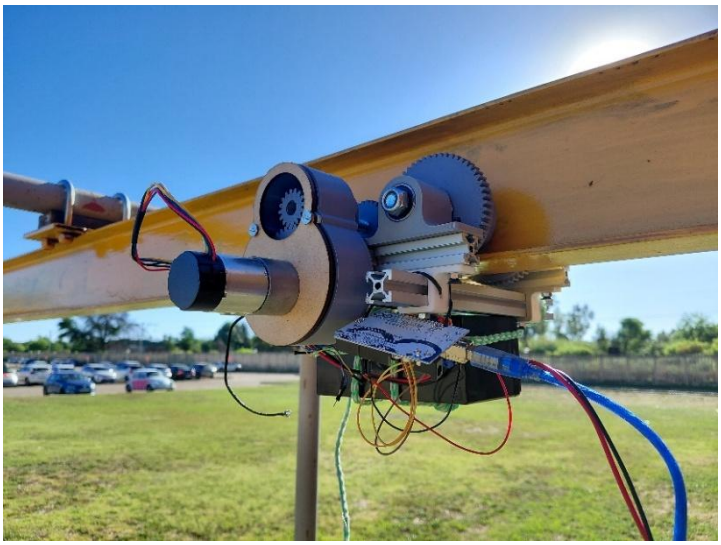


Figure 4: MCTR1 motor testing rig as Implementation evidence.

5.2 MCTR 2 - Spooling

Approximately 90% of gears have been printed. To the exception is the combined bevel/herringbone gear connects the motor to the herringbone gear stage. Its completion was delayed by an unplanned power outage at the fabrication lab which caused failed prints and added to the backlog of scheduled prints. The gearbox frame is fully built and assembled, although it may be still altered to suit system integration or to mount electrical components. Engineering wood panels still needs to be obtained from the fabrication lab to close off the sides of the frames and to provide drill mounting pints for the gear shafts. The bearing order has been delayed due to the order being misplaced. The spool is finished and has been wounded up with the 3mm cable.

The image below shows the physical progression of gearbox and spool system:



Figure 5: Progress of Gearbox and spooling system.

The S-type load cell has been obtained, and two eye bolts have been fitted so it can be attached in line between the hoist cable and the magnetic attachment. The load cell still needs to be calibrated. This will be first done on a separate microcontroller with the HX744 amplifier, so its output can be characterised and the resulting read-out-code and calibration factor handed over the REII controller.



Figure 6: S type load cell

5.3 EEII 1 – Drive circuit

The LT spice designs and simulations were not included to this report (refer to previous checkpoint documents).

The preliminary test results of the single-phase version used to switch 15V on the DC bus:

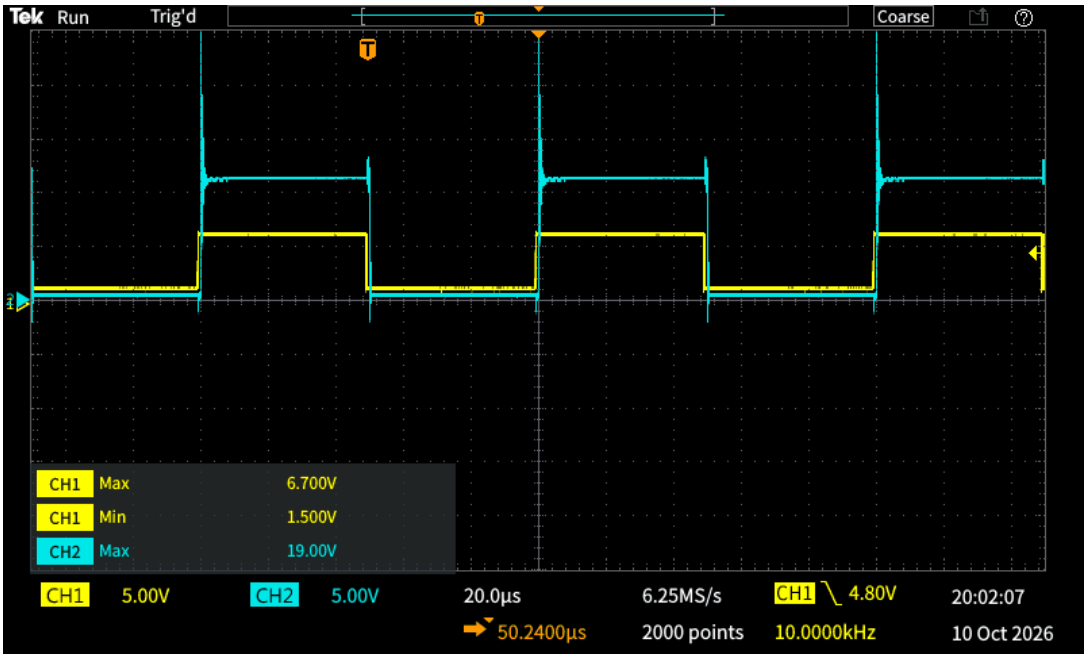


Figure 7: Low voltage single phase switching.

Yellow shows the PWM input at the opto- coupler, blue shows the measured output on the IGBT's.

The schematic of the drive circuit was created in Kicad as in the following figure:

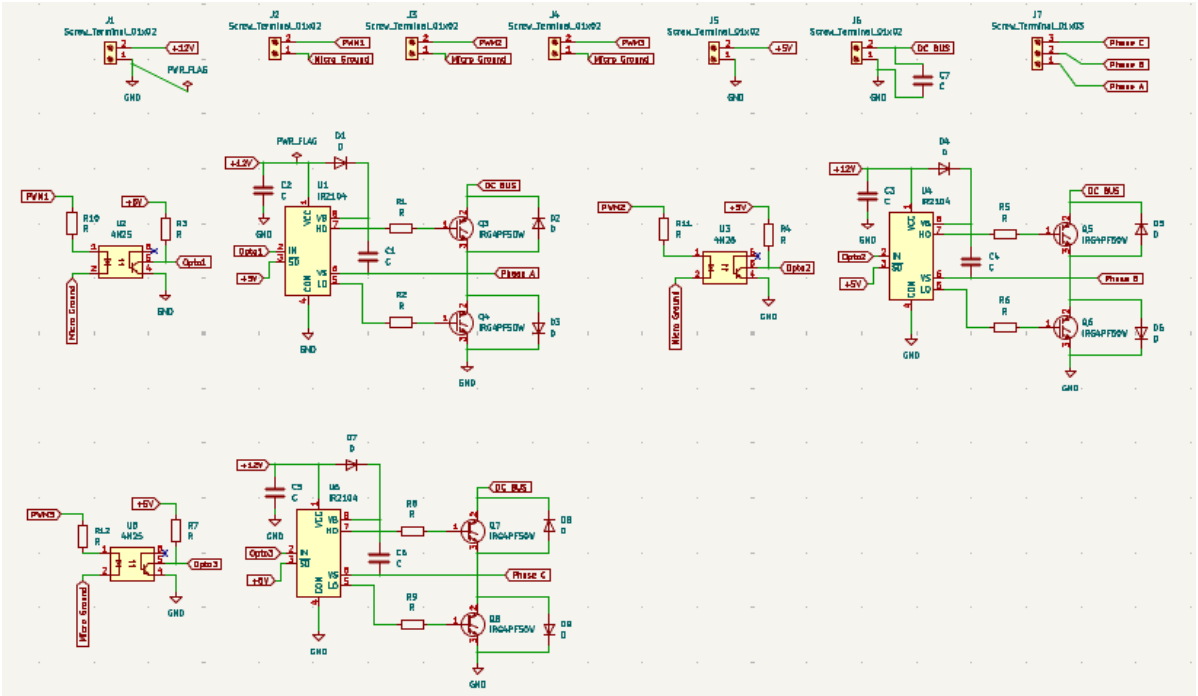
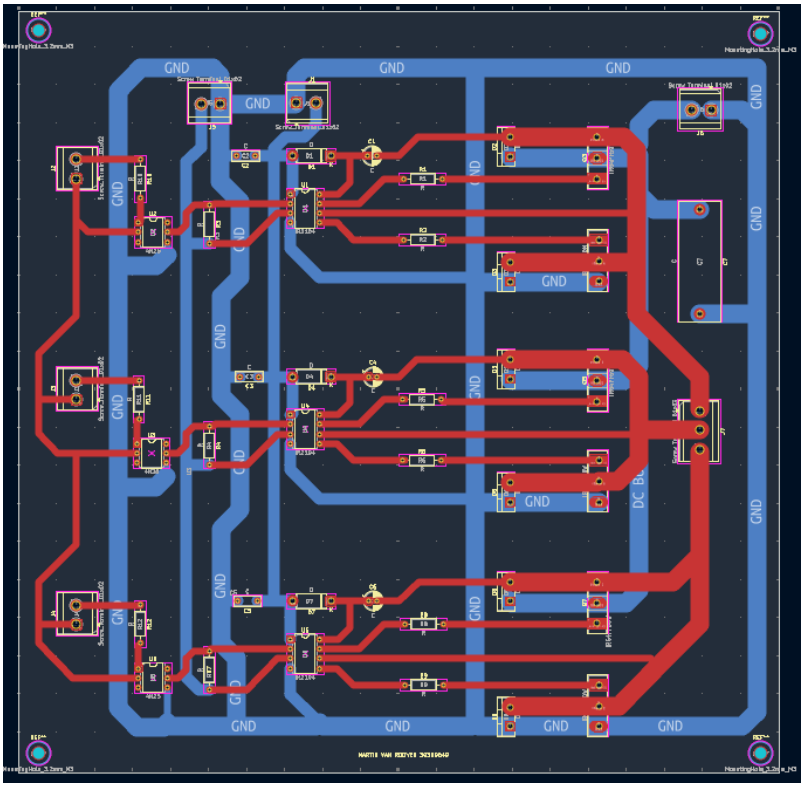


Figure 8: Drive circuit schematic in Kicad

The footprints of each component were added to the schematic, and the PCB design was completed as follows:


GND
GND
GND

5.4 EEII 2 – Magnet

The theoretical design of a core was successful. The cores need to be constructed, and the physical response needs to be tested. The circuit was designed to be mathematically sound, but LTspice simulations need to be set up to verify the theoretical validity of the design. Logic level controller, Buck converter, and current sensor each need to be obtained that meet the circuit requirements. The remaining work is to analyse the circuits, physically construct the cores, and integrate with each of the subsystems the electromagnet communicates with. The core details are seen as:

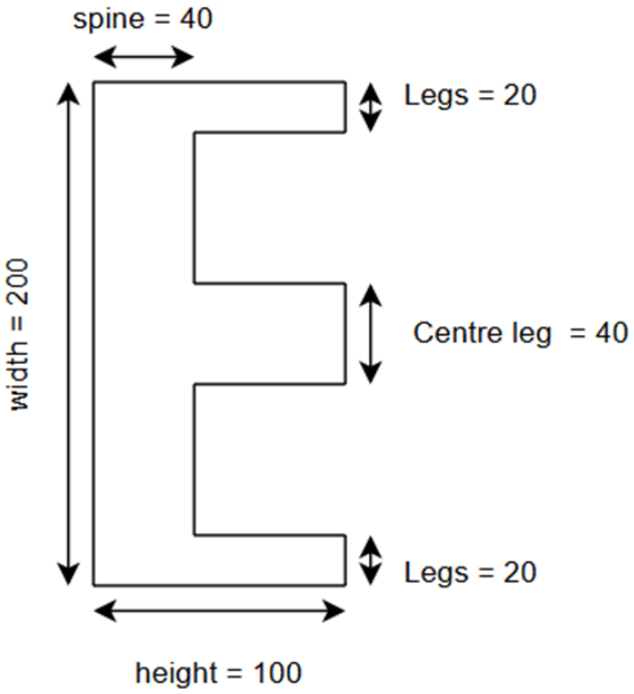


Figure 11: Core details

With the windings around the central leg, the cross sectional area as mentioned is 1600mm^2. The outer legs have half of this area to ensure that the total positive and negative pole faces remain the same. The circuit layout is shown below.

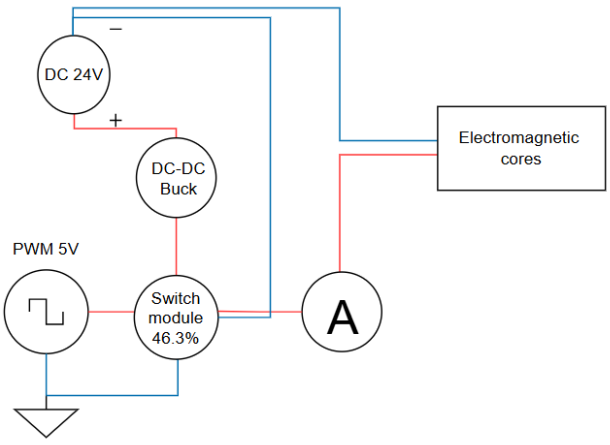


Figure 12: Electromagnet circuit layout.

With these being theoretically correct, the tests and simulations can be conducted as well as the case design for the electromagnets can be started.

Currently, the switching circuit was tested as a proof of concept to verify the response as calculated.



Figure 13: Magnet switching circuit testing.

The results align with the predicted response; this setup used a 5V input at a duty cycle of 40% and the resulting voltage was seen as 2V. This will be how the microcontroller adjusts the DC input from the supply to the magnetic coils.

The cores each work in parallel as described in the operation of the electromagnet, but due to high current and temperature generation, the cores will need housing. This housing also helps to balance the payload and hoist connection. The current design of the housing is as follows.

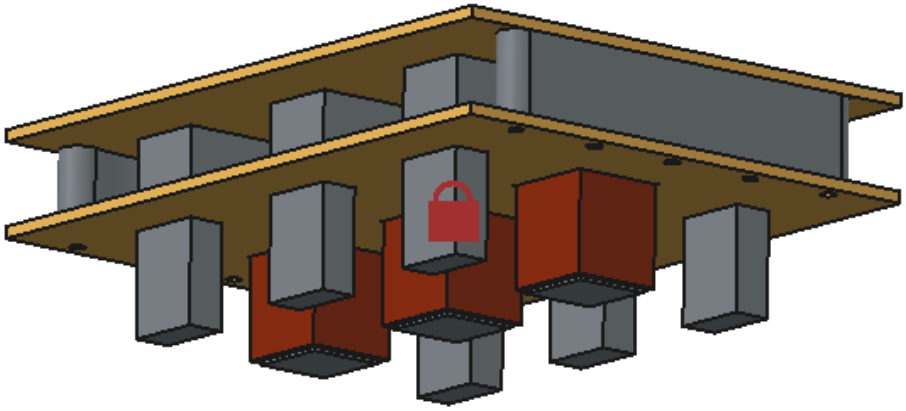


Figure 14: Magnet housing design

The housing consists of two sheets holding the cores in place. The housing is separated with two 3d printed spacers which have screw holes to tighten and equally spread the load force across the structure. The housing without the cores seen from the bottom.

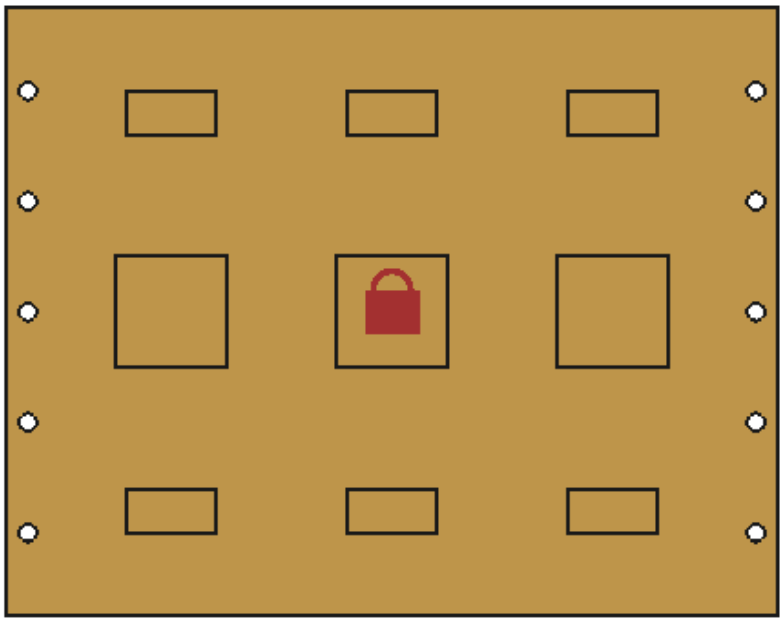


Figure 15: Magnet housing top view

The spacing for the central legs was designed to be bigger than necessary to avoid damaging the windings. The outer legs were designed to fit tightly to keep the cores aligned and in place.

5.5 REII- Remote Control Dashboard and RF Ground Station

The REII subsystem has completed wireless range and interference testing, finalized the complete pin wiring map for the shared ESP32-S3 main board across all five group members, written and bench-tested the wireless firmware on both the NodeMCU-32S ground station and ESP32-S3 crane microcontrollers, and verified live two-way communication at the final operating settings (Channel 77 / 2477 MHz, 2 Mbps, 32-byte static payloads). Table 44 summarizes the REII milestones and measured implementation evidence.

Table 44: REII milestones and implementation evidence.

REII Milestone / Deliverable	Status	Measured Implementation Evidence & Key Results
1. Wireless Radio Bring-Up & 1–190 m Range Test	Completed & Verified	Tested 1000-packet bursts from 1 m to 190 m across all three data speeds. Confirmed that 10 µF + 100 nF capacitors across the radio power pins stop voltage drops and low power on the bench (< 2 m) prevents close-range overload, achieving 0.0% packet loss across the 1–10 m crane operating zone.

2. 2.4 GHz Wi-Fi Scan & Channel Plan	Completed & Locked	Scanned all 126 channels in the 2.4 GHz band and found 17.2–22.3% packet loss on campus Wi-Fi channels (Channels 2, 6, and 11) vs. 0% interference on Channel 77 (2477 MHz). Locked Channel 77 at 2 Mbps and set up a 5-group channel plan [4, 50, 73, 77, 81].
3. Main Board Schematic Check & J1–J9 Pin Map	Completed & Locked	Verified all 5 sheets of the ESP32-S3 carrier board schematic and assigned every group member's pins: REII radio on Header J1 (J1.1–J1.6, J1.8) and J3.3; ON/OFF signals on Headers J3 & J4 (EEII 1 on J3.4–J3.5, EEII 2 on J3.6, MCTR 1 on J4.4–J4.7, MCTR 2 brake on J4.8); PWM/encoders on Header J7 (J7.5–J7.9, J7.12); and analog sensors on Header J6 (J6.5–J6.8).
4. Two-Way Radio Software & 200-Packet Live Test	Implemented & Bench-Tested	Programmed the NodeMCU-32S ground station (COM10) and ESP32-S3 crane board (COM12) with nRF24L01+ radios (isPVariant=1, 4 MHz SPI, 32-byte payload, CRC-16, Auto-ACK, pipes 'REII2'/'REII1'). On Channel 77 at 2 Mbps (PA_MAX, ARC=5, ARD=500 µs), a live 200-packet test (smoke.csv) achieved 200/200 forward packets received on the crane (rig.log frames #5–#204, 100.0% uplink delivery), 192/200 Auto-ACKs (96.0% first-try confirmation, 4.0% PER < 5.0% limit), and 0.89 ms mean RTT (0 ms min, 2.0 ms max on Round #142, well within the 4.15 ms limit) while concurrently polling the 24-bit ADS1220 ADC (0.0041–0.0042 V) and AHT20 (32.35 °C) as given by the board starter kit.
5. 32-Byte Data Packets, PC Dashboard & Anti-Sway	In Progress (Ready for Rail Integration)	Completed the 32-byte command (0xA5) and sensor reply (0x81) packet code (including EEII 2 magnet current on Byte 14 / pin J6.8), the Python PC control dashboard, and the smooth anti-sway speed ramp calculator ahead of full rail testing. And clarifying obstacle avoidance implementation as requested.

5.5.1 Dual-Node Hardware Bring-Up and RF Register Verification (Milestone 4 Evidence)

To verify Milestone 4 prior to rail mounting, an automated dual-serial bench test (logged 2026-10-16 at 115 200 bit/s) was executed across the NodeMCU-32S USB ground station (COM10) and the shared ESP32-S3 crane System Controller (COM12, FU8). As recorded in the archived ground-station serial capture (groundstation.log), the ground station confirmed USB-CDC echoing (>>> ECHO bringup_evidence -> ECHO,bringup_evidence in 6 ms) and verified hardware registers (radio.begin() OK, isPVariant=1, 4 MHz SPI clock, 16-bit CRC, 32-byte static payloads, Auto-ACK enabled, and 5-byte addresses 0x3249494552 / 'REII2' and 0x3149494552 / 'REII1'). Sending >>> APP switched the link from diagnostic Channel 80 (250 kbps) to the crane profile (OK,app_ch=77,2M,PA_MAX,ARC5,ARD500). Figure 16 shows the bench setup of the crane MCU and USB ground-station transceivers.

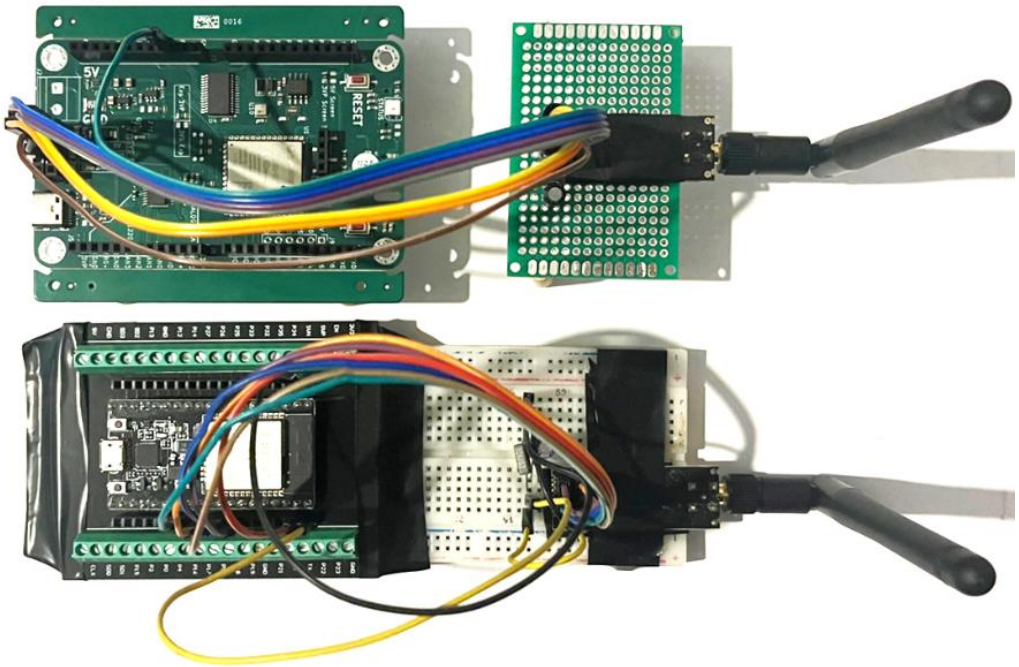


Figure 16: Bench test setup of the ESP32-S3 crane System Controller (FU8, COM12) and prototype NodeMCU-32S USB ground-station radio unit (FU6, COM10) during Milestone 4 two-way RF verification.

5.5.2 200-Packet Live Smoke Test (smoke.csv)

With both radios locked on Channel 77 (2477 MHz) at 2 Mbps, the ground station transmitted 200 back-to-back 32-byte PING packets (~72 ms period) from 12:54:40.421 to 12:55:18.804 UTC. Cross-referencing the per-round smoke test dataset (smoke.csv, plotted in Figure 17) against the archived timestamped serial captures (groundstation.log, rig.log and index.json) confirms three key implementation results:

- **100.0% Forward Uplink Delivery (200 / 200 Received on Crane Rig):** In rig.log, the crane controller's nRF24 frame counter stood at frames=4 prior to the burst (12:54:35.350) and logged every incoming packet sequentially from PING #5 (12:54:40.411) through PING #204 (12:55:15.791), confirming 200/200 forward packets received (0.0% forward command loss).
- **96.0% Auto-ACK Confirmation & Close-Range PA_MAX Diagnosis:** In smoke.csv, 192/200 packets returned an immediate hardware Auto-ACK (4.0% PER, passing the < 5.0% target). On the 8 timeout rounds (#18, #96, #97, #130, #152, #160, #190, #200), rig.log confirms the crane received the forward PING every time (e.g., Round #18 logged as PING #22 at 12:54:41.631), proving that only the return ACK dropped due to close-range (< 1 m) LNA saturation at PA_MAX—validating our ICD rule to use PA_LOW under 2 m.
- **0.89 ms Mean Round-Trip Latency (2.0 ms Max vs. 4.15 ms Limit):** Across the 192 ACKed rounds in smoke.csv, 22 packets (11.0%) completed in 0 ms (< 0.5 ms), 169 packets (84.5%) in

1 ms, and 1 packet (Round #142, 0.5%) in 2 ms after one retry—giving a 4.6× margin below the 4.15 ms limit.

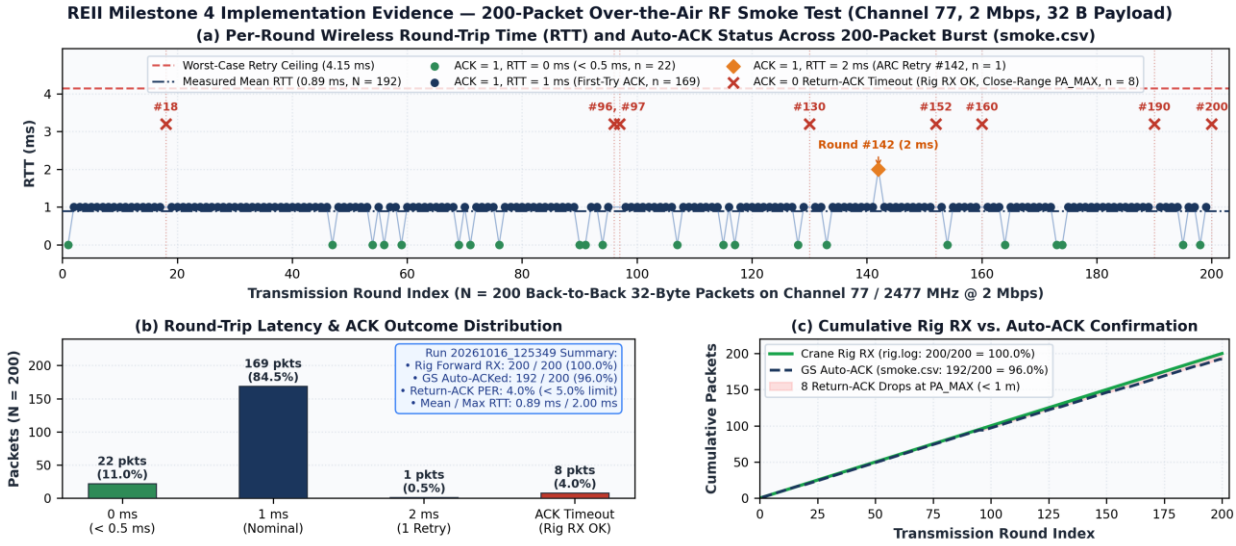


Figure 17: Measured 200-packet wireless smoke test results (smoke.csv, Run 20261016) on Channel 77 (2477 MHz) at 2 Mbps: (a) per-round RTT and Auto-ACK status; (b) latency and outcome distribution (0.89 ms mean RTT); and (c) cumulative crane rig reception (200/200) vs. ground-station Auto-ACK confirmation (192/200).

Appendix A

Table 45: Acronyms and abbreviations.

Term	Definition
[Term]	[Definition]
UART	Universal Asynchronous Receiver-Transmitter
CAN	Controller Area Network
JST	Japan Solderless Terminal
PSU	Power Supply Unit
TTL	Transistor to Transistor Logic
CMOS	Complementary Metal-oxide Semiconductor
ARC / ARD	Auto-Retry Count (0–15 retries) and Auto-Retry Delay (250–4000 µs)
ESB	Enhanced ShockBurst hardware packet engine on nRF24L01+
RPD	Received Power Detector (> −64 dBm in-band RF energy indicator)
ADC	Analog-to-Digital Converter (converts sensor voltages into digital numbers)
ACK / Auto-ACK	Automatic confirmation packet sent back by the crane radio when a command arrives
CE / CSN	Chip Enable (radio listen/transmit pin) and Chip Select Not (SPI device select pin)
CMOS / TTL	Standard 3.3 V / 5 V digital logic voltage levels
CRC-16	16-bit error-check code attached to every wireless packet to detect corrupted data
GPIO	General Purpose Input / Output pin on a microcontroller
I/O	Input / Output
I2C	Two-wire serial bus used on the main board to control the digital pin expander chip
MCU	Microcontroller Unit
PER / RTT	Packet Error Rate (% of lost packets) and Round-Trip Time (ms for a command and reply)
PWM / SPWM	Pulse Width Modulation / Sinusoidal Pulse Width Modulation
RF	Radio Frequency (2.4 GHz wireless communication)
SPI	Four-wire Serial Peripheral Interface used between the microcontroller and radio module
TBD / TBC	To Be Determined / To Be Confirmed
VFD	Variable Frequency Drive (controls the three-phase AC induction motor)
ADC	Analog-to-Digital Converter (converts sensor voltages into digital numbers)